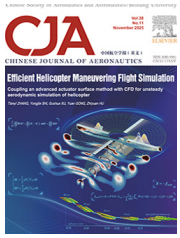
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REVIEW

Advancements in incremental nonlinear dynamic inversion and its components: A survey on INDI – Part II



Agnes STEINERT [a,*], **Stefan RAAB** [a], **Simon HAFNER** [a], **Florian HOLZAPFEL** [a], **Haichao HONG** [b]

[a] *Institute of Flight System Dynamics, Technical University of Munich, Garching near Munich 85748, Germany*
[b] *School of Aeronautics and Astronautics, Shanghai Jiao Tong University, Shanghai 200240, China*

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Abstract Incremental Nonlinear Dynamic Inversion (INDI) is a control approach that has gained popularity in flight control over the past decade. Besides the INDI law, several common additional components complement an INDI-based controller. This paper, the second part of a two-part series of surveys on INDI, aims to summarize the modern trends in INDI and its related components. Besides a comprehensive components specification, it addresses their most common challenges, compares different variants, and discusses proposed advances. Further important aspects of INDI are gain design, stability, and robustness. This paper also provides an overview of research conducted concerning these aspects. This paper is written in a tutorial style to familiarize researchers with the essential specifics and pitfalls of INDI and its components. At the same time, it can also serve as a reference for readers already familiar with INDI.

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1. Introduction

Incremental Nonlinear Dynamic Inversion (INDI) is a rapidly evolving control methodology - while in 2016, around 180 publications on the topic were available, by 2020, the number rose to more than 600, and by April of 2024 to more than 1 500, according to Google Scholar. Mainly being investigated and developed within the flight control community, in recent years INDI spread over to other domains and was applied to spacecraft,[1,2] an unmanned underwater vehicle,[3] pressurized water reactor-based small modular reactors,[4] and a pneumatic actuation system.[5]

* Corresponding author.
E-mail addresses: agnes.steinert@tum.de (A. STEINERT), stefan.raab@tum.de (S. RAAB), simon.hafner@tum.de (S. HAFNER), florian.holzafpel@tum.de (F. HOLZAPFEL), haichao.hong@sjtu.edu.cn (H. HONG).

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INDI evolved in the flight control domain as an alternative to Nonlinear Dynamic Inversion (NDI), which is a popular flight control methodology that has been applied, especially in the 1990s and 2000s, to several aircraft.[6–8] NDI is a model-based approach that depends on the quality of the mathematical model utilized. Ref. 9 encountered robustness issues when the model's uncertainty was too large. As a consequence, INDI emerged with the characteristics of being less model-dependent. INDI is a sensor-based approach that replaces some model dependencies with appropriate measurements. Researchers found this replacement especially beneficial for novel aircraft configurations, such as transition vehicles, where exact model knowledge is lacking or too costly to identify. Besides that, the advantage that INDI can address input non-affine systems makes it suitable for overactuated systems with input nonlinearities such as for example aerial vehicles with tilting rotors or wings. Furthermore, INDI significantly improves disturbance rejection compared to NDI because the measurements in contrast to a model contain information about disturbances acting on the vehicle. Consequently, an INDI-based controller rejects disturbances one integration level earlier than NDI. As mentioned, INDI also started to draw interest in various other fields. As these areas start to investigate the potential of INDI, it has become evident that a detailed overview summarizing its developments and challenges is necessary. Over the last decade, our institute has engaged in numerous projects and collaborated with other INDI researchers through conferences and joint projects. These experiences have provided a thorough overview of INDI's diverse applications and advancements.

In the first part of this two-part series of surveys on INDI, Ref. 10 summarized different derivations of INDI found in the literature, depending, for example, on whether the system is input affine or not. It gives an overview of the most commonly used variants of INDI with a tutorial nature and addresses relevant aspects of INDI. Furthermore, it summarizes the evolution of several INDI variants and their characteristics, as the different approaches provide distinct insights and contribute to a better and holistic understanding of INDI. Because INDI faces various challenges, which different research groups discovered, specific advances to the baseline INDI were developed. These advances are the focus of this second part of the survey. Furthermore, an INDI controller consists of several additional components besides the INDI law. Challenges were discovered for each component, and solutions have been developed over time. This paper aims to summarize these components, their purpose, and modern trends.

Additionally, we want to give an overview of different Gain Design approaches for INDI and research conducted regarding its stability and robustness.

The remainder of the paper is set up as follows: Section 2 introduces a basic INDI architecture that, in the authors' view, represents a fundamental structure that contains the main components. This architecture will serve as a baseline to set further advances and modifications in the right context. Section 3 derives INDI laws. Section 4 deals with obtaining pseudo control commands and Section 5 deals with the challenge of estimating the current control inputs. The similar but potentially more difficult challenge of obtaining a proper estimation of the current pseudo control is discussed in Section 6, followed by the important concept of pseudo control/-control input synchronization in Section 7. Section 8 provides a brief overview of control allocation in the context of INDI. Due to the broad scope of control allocation, we will not cover it in detail here. Instead, we will outline some specifics related to INDI. Subsequently, Section 9 deals with assessing the stability and robustness of INDI-controlled systems, followed by Section 10, in which gain tuning methods applied to INDI controllers are discussed. Finally, Section 11 concludes the paper.

2. Components of INDI controllers

INDI controllers are, in general, composed of multiple components as described in Ref. 11. Those components are very similar, often even equal, to those commonly used in NDI laws.[12] One possible structure is visualized in Fig. 1, which contains the following main controller components: (A) Reference model, (B) Error controller, (C) Estimation module, (D) Control allocation.

The following describes this basic architecture to serve as a common baseline: The Reference Model provides reference trajectories in the controlled variables $\boldsymbol{y}$ and its derivatives, up to the reference pseudo control $\boldsymbol{\nu}$. The reference trajectory, denoted as $\boldsymbol{y}_{\text{ref}}, \dot{\boldsymbol{y}}_{\text{ref}}$, and higher-order derivatives, is fed into the Error Controller, where these variables are pairwise compared to the actual plant's trajectory variables, such as $\boldsymbol{y}, \dot{\boldsymbol{y}}$, and corresponding derivatives.

The Error controller then calculates a feedback portion of the pseudo control, named $\boldsymbol{\nu}_{\text{ec}}$, to "pull" the plant trajectory towards the reference trajectory in case of deviations. The reference model also outputs a feedforward signal. Often, this feedforward corresponds to the reference pseudo control $\boldsymbol{\nu}_{\text{ref}}$. Both feedforward and feedback pseudo controls are added together to form a resulting desired pseudo control demand $\boldsymbol{\nu}_{\text{des}}$.

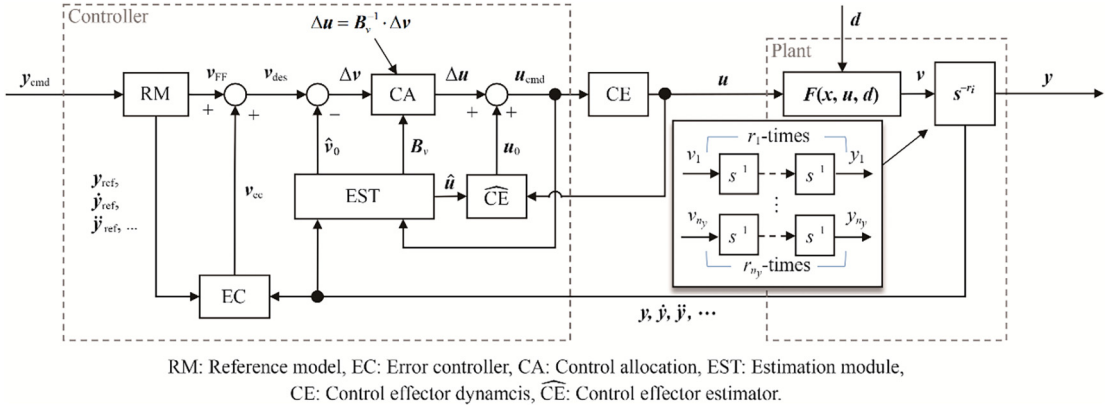
The Estimation module uses measurements and sometimes onboard models (also denoted OBM for Onboard Model [13–16] or OBPM for Onboard Plant Model [11,17,18]) to estimate the current plant pseudo control $\hat{\boldsymbol{\nu}}_0$, which is then subtracted from the desired pseudo control $\boldsymbol{\nu}_{\text{des}}$, to form the desired pseudo control increment $\Delta\boldsymbol{\nu}$.

This increment is fed into the control allocation, along with an estimation of the current control effectiveness $\boldsymbol{B}_{\nu}$ from the Estimation Module. Depending on the application, the control effectiveness can be constant, scheduled over the envelope (e.g., depending on speed and height), and sometimes estimated online. The control allocation calculates the required increment in control inputs $\Delta\boldsymbol{u}$, which should lead to the desired plant response $\Delta\boldsymbol{\nu}$.

The calculated $\Delta\boldsymbol{u}$ is then added to the current control input value $\boldsymbol{u}_0$ that acts on the plant in order to calculate the new control input command $\boldsymbol{u}_{\text{cmd}}$. In this context, the current plant input $\boldsymbol{u}_0$ can be obtained through estimation, measurement, or a combination of both.

Several variants of the described architecture exist, with various modifications to its components. The corresponding sections of this paper will discuss these adjustments and variants. Below is a summary of the main streams and differences to provide context and an outline:

(1) Different variants and types of reference models evolved and were applied to INDI controllers. Section 4 provides a detailed comparison. Besides that, in several applica-



RM: Reference model, EC: Error controller, CA: Control allocation, EST: Estimation module, CE: Control effector dynamcis, $\widehat{CE}$: Control effector estimator.

Fig. 1 Block diagram of a generic INDI controller with usual components reused from Ref. 10.

tions, outer control loops provide the reference commands directly, as will be discussed in Section 4.1. In these cases, no reference model is used.

(2) In many applications, the INDI controller needs some feedback signals, which must be estimated because the related sensors are too expensive or unavailable. For example, an onboard plant model might estimate the control effectiveness. Additionally, we might need to measure or estimate the current pseudo control $\boldsymbol{v}_0$ or the current input $\boldsymbol{u}_0$. Different possibilities to obtain $\boldsymbol{u}_0$ are addressed in Section 5, different possibilities to obtain $\boldsymbol{v}_0$ in Section 6.

(3) In aircraft applications, the pseudo control $\boldsymbol{v}_0$ is commonly the angular acceleration, for which direct sensors are primarily unavailable. Several approaches evolved to obtain the angular accelerations. Some of them introduce new challenges and different solutions developed to deal with those associated issues. Sections 6, 7 will discuss these challenges and solutions.

(4) As will be addressed in Section 8, the control allocation allocates, in the context of INDI, often the desired pseudo control increments to effector command increments. For over-actuated systems, additional constraints can be imposed in the control allocation. Additionally, allocation strategies might account for effector and rate saturations. Control allocation is a wide field of research, and we will only address some aspects that we found worth mentioning in the context of INDI.

In the next section, different INDI laws are derived and presented.

3. Overview of sensor-based INDI and INDI with actuators

There exist different variants of INDI. The most common variants are model-based INDI, sensor-based INDI [19–30] and hybrid INDI.[16,25,31,32] This publication is the second part of a two-part survey. The first part of this two-part series of surveys on INDI is presented in Ref. 10. There, we discuss various variants of INDI and their derivation in a tutorial style. Depending on different aspects, for example, whether the system is input affine or not, the INDI laws can be derived in different ways. If the system is input affine, then the INDI law can be obtained via direct inversion.[10] For input non-affine systems, the input nonlinearity can be addressed by local linearization.

The following section derives sensor-based INDI for input non-affine systems. The section afterwards gives a brief overview of two INDI variants that actively account for actuator dynamics. For details, however, Ref. 10 is recommended.

3.1. Sensor-based INDI for input non-affine systems

In order to establish a common basis, we repeat here the derivation of sensor-based INDI that from our point of view offers intuitive understanding of INDI, and which is deduced from various ways of derivation found in literature.[9,11,32–34] The full derivation can be found in the first paper of this two series review on INDI (Part I).[10] In the following, we simplify the derivation to a single-state, input non-affine Single Input Single Output (SISO) system given by

$$\begin{cases} \dot{x} &= f(x) + g(x,u) \\ y &= h(x) \end{cases} \tag{1}$$

with a relative degree of one such that

$$\dot{y} = \frac{\partial h(x)}{\partial x}\dot{x} = a_v(x) + b_v(x,u) = F(x,u) \tag{2}$$

with state-only dependent part $a_v(x) = \frac{\partial h(x)}{\partial x} f(x)$ and control input-dependent part $b_v(x,u) = \frac{\partial h(x)}{\partial x} g(x,u)$ being non-zero. Note that the derivation is generalized to non-scalar MIMO systems with arbitrary relative degree r in the first part of our survey in Ref. 10, as well as in Refs. 11,35,36 for example.

We redefine the time derivative $\dot{y}$ as pseudo control v and its dependency on states and inputs as $F(x,u)$,

$$v = F(x,u) = \dot{y} \tag{3}$$

The redefinition of the pseudo control v serves two purposes: First, it helps to align the nomenclature, because later an "error" or "deviation" in the pseudo control is calculated by subtracting the true plant pseudo control v from a desired pseudo control v_{des}. Second, in MIMO systems where multiple output channels may have different relative degrees, the notation of pseudo control v remains independent of the actual rel-

ative degree, it just represents a vector containing all the responsive time derivatives $y_i^{(r_i)}$ of the different outputs.

A perfect inversion of the nonlinearity given in Eq. (3) could be achieved if $F(x,u)$ would be known exactly and could explicitly be solved for u, resulting in the inversion law

$$u = F^{-1}(x,v) \tag{4}$$

This inversion law can be used to impose a desired v_{des} such that

$$\begin{cases} v &= F(x,F^{-1}(x,v_{\text{des}})) \\ v &= v_{\text{des}} \end{cases} \tag{5}$$

The combined system, consisting of the system to be controlled and its inversion, has inputs v_{des} and outputs y. Hence, v are denoted as pseudo controls since they act as new inputs into the combined system. We distinguish v_{des}, which is the desired pseudo control, and v, which is the true system reaction. In case of an imperfect inversion and in the presence of disturbances, v_{des} and v may differ.

In the current operating point (u_0,x_0), which means the physical realization of the variables at a distinctive point in time, the plant pseudo control results in

$$v_0 = F(x_0,u_0) = a_v(x_0) + b_v(x_0,u_0) \tag{6}$$

The control input represent the degree of freedom that we can change in order to achieve a desired pseudo control v_{des}. We denote the ideal control input as u_{ideal}, which would satisfy

$$v_{\text{des}} = F(x_0,u_{\text{ideal}}) = a_v(x_0) + b_v(x_0,u_{\text{ideal}}) \tag{7}$$

It corresponds to the control input u_{ideal} that would be required, to obtain a desired pseudo control v_{des} at the current moment. Subtracting Eq. (6) from Eq. (7) leads to

$$\begin{aligned} v_{\text{des}} - v_0 &= F(x_0,u_{\text{ideal}}) - F(x_0,u_0) \\ &= b_v(x_0,u_{\text{ideal}}) - b_v(x_0,u_0) \end{aligned} \tag{8}$$

where the state-only dependent term $a_v(x_0)$ cancels completely.

In the following, we distinguish two cases:

- **Case 1.** Input affine systems.
- **Case 2.** Input non-affine systems.

If the system is input affine (Case 1), meaning $b_v(x,u) = B_v(x)\cdot u$, Eq. (8) results as a linear equation

$$v_{\text{des}} - v_0 = B_v(x_0)\cdot u_{\text{ideal}} - B_v(x_0)\cdot u_0 \tag{9}$$

for which u_{ideal} can be exactly calculated as

$$u_{\text{cmd}} = B_v^{-1}(x_0)\cdot(v_{\text{des}} - v_0) + u_0 \tag{10}$$

where u_{cmd} corresponds to the ideal control input u_{ideal}.

If the system is input non-affine (Case 2), one could try to solve the nonlinear equation

$$v_{\text{des}} - v_0 = b_v(x_0,u_{\text{ideal}}) - b_v(x_0,u_0) \tag{11}$$

for u_{ideal} using iterative methods or other approaches. This might be difficult in practice, especially for overactuated systems. Therefore, the nonlinear function $b_v(x_0,u)$ is approximated by its linearization around u_0, given by

$$\tilde{b}_v(x_0,u) = b_v(x_0,u_0) + \tfrac{\partial b_v(x_0,u)}{\partial u}\cdot|_{u=u_0}\cdot(u-u_0) \tag{12}$$

similar to the derivation of Sensory NDI of Refs. 34,37.

Due to the approximation we will not obtain an exact solution anymore. Hence, we relabel u_{ideal} to a "required" u_{req}, which, according to the approximated equation, will lead to the desired pseudo control. Inserting the approximation Eq. (12) in Eq. (11) results in

$$\begin{aligned} v_{\text{des}} - v_0 &= \tilde{b}_v(x_0,u_{\text{req}}) - b_v(x_0,u_0) \\ &= \tfrac{\partial b_v(x_0,u)}{\partial u}\cdot|_{u=u_0}\cdot(u_{\text{req}} - u_0) \end{aligned} \tag{13}$$

Substituting the partial derivative by

$$B_v(x_0,u_0) = \tfrac{\partial b_v(x_0,u)}{\partial u}\cdot|_{u=u_0} \tag{14}$$

results in an equation that is similar to the input affine Eq. (8)

$$v_{\text{des}} - v_0 = B_v(x_0,u_0)\cdot(u_{\text{req}} - u_0) \tag{15}$$

The resulting inversion law is

$$u_{\text{cmd}} = B_v^{-1}(x_0,u_0)\cdot(v_{\text{des}} - v_0) + u_0 \tag{16}$$

where the required control input u_{req} has been renamed to u_{cmd}. $B_v(x_0,u_0)$ is the current sensitivity of the pseudo control with respect to the control inputs, which may depend on current states and/or control inputs. It is denoted as "Control Effectiveness". The onboard Estimation module can provide the latter, either by pre-calculating it for different operating conditions and storing it in a scheduling table or by determining it online from a nonlinear onboard model, which is linearized at the current operating point.[11,13,14]

The terms u_0 and v_0 are measurements and/or estimates of the current control input and pseudo control value.

We will address each of the other entities in Eq. (16) in its own section: Section 4 deals with obtaining the pseudo control demand v_{des}, Section 5 with obtaining the current control input u_0, and Section 6 with obtaining the current pseudo control estimate $v_0 = \dot{y}_0$.

For overactuated systems, various control allocation methods can be used, which we will detail in Section 8. For instance, one possibility is to apply the pseudo inverse.[38]

3.2. Extended INDI (E-INDI) and Actuator NDI (ANDI) – Consideration of actuator dynamics

When deriving the INDI control law, usually actuator dynamics are neglected, or the commanded u_{cmd} is assumed to be obtained instantaneously. The control allocation calculates the required input increment Δu-based on the desired pseudo control increment Δv, for instance, a desired moment increment (In fact, for INDI, it is more common to have "angular accelerations" as pseudo controls for the rotational controls. They are a kinematic entity whose estimation can be supported through available gyros, which measure the angular rates. Moments, on the other hand, are more difficult to estimate or measure. However, since angular accelerations and moments algebraically are on the same level and moments have a more physical intuition, we often speak of moments instead of angular accelerations). Due to actuator dynamics, the desired and actual produced moments will deteriorate in phase and gain as visualized by Ref. 39 in the context of an INDI law. Further challenges might arise in case multiple effectors exist with different dynamics. This is, for example, the case for tilting rotors, where the thrust dynamics are faster than the tilt dynamics. To compensate for these effects, Ref. 39

proposed to multiply the control allocation output with specific gains, which they derive for a first-order and second-order actuator.

Ref. 40 proposed a continuous time formulation of the INDI law in which they can incorporate the actuator dynamics to compensate for their influence, which shall be elaborated in the following in more detail under the name "Extended INDI" (E-INDI). Considering a MIMO system where the pseudo control derivative can be approximated by

$$\dot{\boldsymbol{\nu}} = \boldsymbol{A}_{\nu}(\boldsymbol{x}) + \boldsymbol{B}_{\nu}(\boldsymbol{x},\boldsymbol{u}) \cdot \dot{\boldsymbol{u}} \approx \boldsymbol{B}_{\nu}(\boldsymbol{x},\boldsymbol{u}) \cdot \dot{\boldsymbol{u}} \tag{17}$$

with $\boldsymbol{x} \in \mathbb{R}^{n_x}, \boldsymbol{\nu} \in \mathbb{R}^{n_y}, \boldsymbol{u} \in \mathbb{R}^{n_u}$. For $n_y = n_u$, the inversion law of E-INDI is given by

$$\begin{aligned} \boldsymbol{u}_{\text{cmd}} &= (\boldsymbol{B}_{\nu}(\boldsymbol{x},\boldsymbol{u}) \cdot \omega_{\text{OL}})^{-1} \cdot \dot{\nu}_{\text{des}} + \boldsymbol{u}_0 \\ &= (\boldsymbol{B}_{\nu}(\boldsymbol{x},\boldsymbol{u}) \cdot \omega_{\text{OL}})^{-1} \cdot \omega_{\nu,\text{des}}(\nu_{\text{des}} - \nu) + \boldsymbol{u}_0 \end{aligned} \tag{18}$$

where $\dot{\nu}_{\text{des}} = \omega_{\nu,\text{des}}(\nu_{\text{des}} - \nu)$ are desired first-order pseudo control dynamics. For MIMO systems both ω_{OL} and $\omega_{\nu,\text{des}}$ are diagonal matrices, where the diagonal elements represent the bandwidth of the respective channels. If the actuator is a linear first-order system then $\omega_{\text{OL}} = 1/T$, where T is the time constant. In case of linear second-order actuator dynamics of the form $\frac{\omega_0^2}{s^2+2\zeta\omega_0 s+\omega_0^2}$, ω_{OL} is specified by $\omega_0/(2\zeta)$.

As visible in Eq. (17), an alternative way is to directly command a desired $\dot{\nu}$ to the control allocation, which allocates a necessary $\dot{\boldsymbol{u}}$, which in subsequent steps is translated into a required $\boldsymbol{u}_{\text{cmd}}$. This is beneficial if the control effector dynamics are nonlinear. For example the dynamics of an electric motor driving a rotor, might be described by

$$\dot{\omega}_p = K_1\omega_p^2 + K_2\omega_p + K_3\omega_{\text{cmd}} \tag{19}$$

A desired $\dot{\nu}$ can be allocated into a necessary $\dot{\omega}_p$. Then the relation Eq. (19) might be inverted for the actual command to be sent to the motor, ω_{cmd}.

Another example where an allocation on the $\dot{\nu}$ level was useful can be found in Ref. 41. Here, the relation between pseudo control and input is quadratic $\nu = B_{\nu} \cdot \omega^2$. Defining the virtual control $u = \omega^2$, a desired $\dot{\nu}$ is allocated into $\dot{u}$. Then via the relation $\dot{u} = 2\omega\dot{\omega}$ the necessary $\dot{\omega}$ is calculated. Finally by inverting the actuator dynamics given by $\dot{\omega} = \omega_{\text{OL}}(\omega_{\text{cmd}} - \omega)$ the necessary command ω_{cmd} is obtained.

More details and a comprehensive summary and derivation of E-INDI can be found in Ref. 10.

Later, in 2023, Ref. 42 further investigated the concept of an extended INDI law, using NDI with included actuator dynamics. Their analysis and derivations provide several insights. They show analytically that the standard INDI control law will approach the NDI law for high actuator bandwidth. For further details, please refer to Refs. 10,42. For reference in later sections we repeat the derivation of ANDI in the following. For simplicity, we will consider in the following a single-state SISO system with relative degree of $r = 1$, given by

$$\begin{cases} \dot{x} &= f(x,u) \\ y &= h(x) \end{cases} \tag{20}$$

where x is the system state, u is the actuator state, y is the output. Then

$$\dot{y} = \frac{\partial h(x)}{\partial x}\dot{x} = \frac{\partial h(x)}{\partial x} f(x,u) = F(x,u) \tag{21}$$

and

$$\begin{aligned} \ddot{y} &= \tfrac{\partial F(x,u)}{\partial x}\dot{x} + \tfrac{\partial F(x,u)}{\partial u}\dot{u} \\ &= F_x(x,u)\dot{x} + F_u(x,u)\dot{u} \end{aligned} \tag{22}$$

Assuming linear first-order actuators given by $\dot{u} = \omega_a(u_{\text{cmd}} - u)$ and inserting into Eq. (22) together with the pseudo control $\nu_{\ddot{y},\text{des}} = \ddot{y}$, Eq. (22) can be solved for u_{cmd}, resulting in the NDI law

$$u_{\text{cmd}} = \frac{1}{F_u(x,u)\omega_a}(\nu_{\ddot{y},\text{des}} - F_x(x,u)\dot{x}) + u \tag{23}$$

An interesting observation can be made by choosing the pseudo control based on a special formulation of the error dynamics, as detailed in Ref. 42 and Ref. 10. This special choice will result in the pseudo control

$$\nu_{\ddot{y},\text{des}} = \ddot{y}_{\text{des}} + \omega_e\dot{e} + \omega_a\dot{e} + \omega_e\omega_a e \tag{24}$$

with $e = y_{\text{des}} - y$ and where ω_a defines the actuator dynamics and ω_e specifies slower desired system error dynamics. Inserting this pseudo control into the control law Eq. (23) will result in

$$\begin{aligned} u_{\text{cmd}} &= \tfrac{1}{F_u(x,u)\omega_a}(\ddot{y}_{\text{des}} + \omega_e\dot{e} - F_x(x,u)\dot{x}) \\ &\quad + \tfrac{1}{F_u(x,u)\omega_a}(\omega_a\dot{e} + \omega_a\omega_e e) + u \end{aligned} \tag{25}$$

For infinitely fast actuators, that is infinite bandwidth ω_a, the first term in Eq. (25) becomes zero and we obtain

$$u_c = \frac{1}{F_u(x,u)}(\nu_{\text{des}} - \dot{y}) + u \tag{26}$$

where $\nu_{\text{des}} = \dot{y}_{\text{des}} + \omega_e e$. We compare this result to the classical sensor-based INDI law, Eq. (16), and see that because $B_{\nu}(x_0,u_0) = F_u(x_0,u_0)$ and $\nu_0 = \dot{y}$, both control laws are the same.

The key advantage of the E-INDI and ANDI approaches over classical INDI is that the actuator and pseudo control loops are decoupled. This has the following benefits:

(1) The control effector dynamics can be accounted for in the inversion law.
(2) Multiple control effectors with different bandwidths can be considered this way, addressing the challenge of pseudo control cross couplings introduced by the control effector dynamics (compare Part I of this survey).[10]
(3) For overactuated systems, the challenge that the pseudo control response in classical INDI is a control allocation (and potentially time) dependent combination of control effector dynamics can be addressed (compare Part I of this survey).[10]
(4) For each pseudo control channel the desired dynamics can be specified independently, providing a straightforward way to design outer loops.

4. How to obtain desired pseudo control command – Error controller and reference models

Considering the classical INDI law given in Eq. (16), we want to discuss in this section different approaches described in the literature to obtain the pseudo control command signal ν_{des} and different kinds of reference models used in this context.

The presented methods comprise desired pseudo control based on: (A) Cascaded control loops, (B) Linear reference dynamics, (C) Cascaded physical reference model, (D) Higher order reference model, (E) Incremental physical reference model. The last subsection of Section 4 addresses the topic of envelope protections in reference models. It is worth noting that these methodologies have already been established within the domain of NDI and have been commonly used in Refs. 12,43,44.

Since the primary modification from NDI to INDI replaces the inversion law, additional components such as error controller and reference models can remain unchanged. However, the extension of INDI to E-INDI [40] and ANDI [42] allows additional tailoring of error controller and reference model [45] to the extended INDI law.

4.1. Pseudo control via cascaded control loops

In 2000, Ref. 33 were flying their simplified inversion law on the Vector thrust Aircraft Advanced Control (VAAC) Harrier; for details, see also Ref. 10. The INDI law is only active in the pitch channel, where the pilot commands the pitch rate through a stick input. The desired pitch acceleration $v_{\text{des}} = \dot{q}_{\text{des}}$ for the inversion law is shaped with first-order dynamics following

$$v_{\text{des}} = \dot{y}_{\text{des}} = \omega_d (y_{\text{cmd}} - y_0) \tag{27}$$

where y_{cmd} is the desired pitch rate directly commanded by the pilot, y_0 is the measured pitch rate and ω_d is the bandwidth design parameter. The pseudo control v_{des} from Eq. (27) is used in the INDI law in Eq. (16). As depicted exemplary for a system with $n_u = n_y$ (same number of inputs and outputs) in Fig. 2, the resulting controller resembles a cascaded controller structure with an INDI rate acceleration loop $\dot{y}_{\text{des}} - \dot{y}_0$ and a pitch rate outer-loop $y_{\text{cmd}} - y_0$.

This approach is often used in literature, though with different sources for y_{cmd}. In Ref. 33, the outer-loop y_{cmd} command comes directly from the pilot.

Azinheira,[21] Pfeifle,[24] and Xin [46] use the same structure as Ref. 33. However, with an additional outer-loop, that is, the rates y_{cmd} are not directly commanded by a pilot but generated in an outer attitude control loop. Refs. 46,47 also applied the same approach.

4.2. Pseudo control based on desired error dynamics using linear reference dynamics

Another common approach,[48–50] which is also used commonly in NDI, is to define desired first-order error dynamics according to

$$\dot{e} + \omega_e e = 0 \tag{28}$$

where $e = y_{\text{ref}} - y_0$. With the pseudo control v being $\dot{y}_0$, the error dynamics can be solved for v

$$v_{\text{des}} = \dot{y}_{\text{ref}} + \omega_e (y_{\text{ref}} - y_0) \tag{29}$$

Now different choices for $\dot{y}_{\text{ref}}$ and y_{ref} exist. One common approach, which is for example used in Ref. 49, applies a linear first-order filter - a so-called linear first-order reference model - to generate the reference signals $\dot{y}_{\text{ref}}$ and y_{ref} according to

$$\dot{y}_{\text{ref}} = \omega_{\text{ref}} (y_{\text{cmd}} - y_{\text{ref}}) \tag{30}$$

where y_{cmd} is the pilot command or a command from an outer control loop. The pseudo control is then

$$v_{\text{des}} = \dot{y}_{\text{ref}} + K_{\text{P}} (y_{\text{ref}} - y_0) \tag{31}$$

which corresponds to desired first-order error dynamics given by Eq. (29), as the proportional gain corresponds to the error dynamics bandwidth $K_{\text{P}} = \omega_e$. Fig. 3 depicts the resulting control law.

Sometimes an integral error is added, for example, as presented in Ref. 28

$$v_{\text{des}} = \dot{y}_{\text{ref}} + \left(K_{\text{P}} + \frac{K_{\text{I}}}{s} \right) \left(y_{\text{ref}} - y_0 \right) \tag{32}$$

When using integrators in the error controller, anti-windup strategies must be considered in the presence of actuator limitations. One way to account for such limitations is "Pseudo Control Hedging" (PCH). PCH aims to hide the actuator dynamics from the error controller.[12] Johnson et al.[51–53] developed PCH around the turn of the millennium in the context of neural network adaptive control.

In 2022, Ref. 37 used a linear second-order reference model with natural frequency ω_{ref} and relative damping ζ_{ref} given by

$$\ddot{y}_{\text{ref}} = -2\zeta_{\text{ref}} \omega_{\text{ref}} \dot{y}_{\text{ref}} - \omega_{\text{ref}}^2 y_{\text{ref}} + \omega_{\text{ref}}^2 y_{\text{cmd}} \tag{33}$$

They calculate the pseudo control as follows

$$\begin{aligned} v_{\text{des}} &= \tfrac{K_{\text{I}}}{s} (y_{\text{ref}} - y_0) + K_{\text{P}} (y_{\text{ref}} - y_0) \\ &\quad + K_{\text{D}} (\dot{y}_{\text{ref}} - \dot{y}_0) + K_{\text{FF}} \ddot{y}_{\text{ref}} \end{aligned} \tag{34}$$

In summary, we see that the pseudo control divides into a feedforward part, such as $v_{\text{FF}} = K_{\text{FF}} \ddot{y}_{\text{ref}}$ from Eq. (34) or $v_{\text{FF}} = \dot{y}_{\text{ref}}$ from Eq. (31), and an error controller or feedback part v_{ec}, as also used by Ref. 11. A reference model generates the required reference dynamics: $\dot{y}_{\text{ref}}, y_{\text{ref}}$, etc. and feedforward v_{FF} and an error controller calculates v_{ec} as depicted in Fig. 1.

Other examples for linear reference models in combination with an error controller are presented by Ref. 27 and Ref. 54. Ref. 27 designed control laws for the hover and fixed-wing flight of a quadplane. Both controllers are designed as a cascaded structure with INDI-based inner and outer-loops. The INDI-based control loops are composed of linear reference dynamics and error controllers. The reference models are kept linear, and the error controllers resemble a PID structure. Linear first- or second-order reference models are used depending on the control variables. For the linear first-order reference model along with the error controller, the pseudo control results as

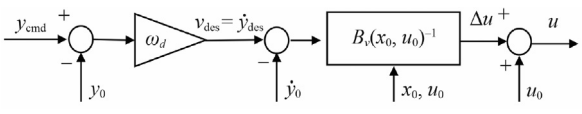


Fig. 2 Cascaded controller with INDI inner-loop.

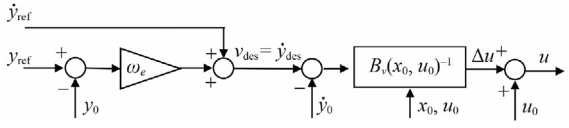


Fig. 3 INDI inner-loop with reference trajectory.

$$\nu_{\text{des}} = \left(K_{\text{P}} + \frac{K_{\text{I}}}{s} + K_{\text{D}}s\right)(y_{\text{ref}} - y_0) \tag{35}$$

This is the same structure that also Ref. 54 uses. For the second-order INDI loop, Ref. 54 used second-order reference dynamics, along with different error controller schemes in different channels, involving various error control laws:

$$\begin{cases} \nu_{\text{des}} &= (K_{\text{P},e} + \frac{K_{\text{I}}}{s})e + (K_{\text{P},\dot{e}} + K_{\text{D}}s)\dot{e} + \ddot{y}_{\text{ref}} \\ \nu_{\text{des}} &= K_{\text{P},e}e + (K_{\text{P},\dot{e}} + K_{\text{D}}s)\dot{e} + \ddot{y}_{\text{ref}} \\ \nu_{\text{des}} &= (K_{\text{P},e} + \frac{K_{\text{I}}}{s})e + K_{\text{P},\dot{e}}\dot{e} + \ddot{y}_{\text{ref}} \end{cases} \tag{36}$$

where $\nu_{\text{des}} = \ddot{y}_{\text{des}}$, and $e = y_{\text{ref}} - y$.

In general, actuator dynamics are not considered when deriving the relative degree of an output. As we have seen (Section 3.2) in E-INDI [40] and ANDI,[42] for MIMO systems with multiple actuators, we can specify equivalent "inner-loop" dynamics of a pseudo control. Furthermore, the same concept can apply when considering cascaded control loops. For example, an attitude controller can be designed on top of an existing angular rate controller. In this case, the inner-loop, consisting of the angular rate controller and the plant, can be viewed as 'actuator dynamics,' with the angular rate command acting as the 'actuator command'. Such a case is considered in the reference model design of Ref. 41 in the context of multicopter roll control. Here, they consider the roll angle Φ as controlled output, which in their case is specified as relative degree one, because they interpret the angular rate $\dot{\Phi}$ as control input that can be commanded. The dynamics of building up $\dot{\Phi}$ is essentially seen as "actuator dynamics" from the Φ control loop and is approximated as first-order linear dynamics. In control theoretical notation, the controlled output is $y = \Phi$ with pseudo control $\nu = \dot{y} = \dot{\Phi}$, which itself has the described equivalent first-order inner-loop dynamics. Rupprecht et al. propose a reference model design that specifically addresses this, such that they employ a **second-order** reference model for the **relative degree one** output $y = \Phi$, as depicted in Fig. 4.[41]

The proposed reference model structure incorporates the following features:

(1) It accounts for the fact that a reference in $\dot{y}$ cannot be achieved instantaneously because of an inner-loop controller and the related first-order inner-loop dynamics.
(2) Saturations are placed on physical entities: it incorporates rate limits $\dot{y}$ and angular acceleration limits $\ddot{y}$ in the reference dynamics.
(3) The feedforward signal includes the angular acceleration limits $\ddot{y}$ and realizes a direct feed-through of command inputs y_{cmd}.

All these features contribute to physically achievable reference trajectories in y. The overall reference dynamics, in the unsaturated case, are given by

$$\begin{aligned} \ddot{y}_{\text{ref}} &= K_2[K_1(y_{\text{cmd}} - y_{\text{ref}}) - \dot{y}_{\text{ref}}] \\ &= -\underbrace{K_1K_2}_{=\omega_0^2}y_{\text{ref}} - \underbrace{K_2}_{=2\zeta\omega_0}\dot{y}_{\text{ref}} + K_1K_2y_{\text{cmd}} \end{aligned} \tag{37}$$

where the natural frequency ω_0 and relative damping ζ are specified over the design parameters K_1 and K_2. The cascaded structure allows the incorporation of the physical limits to result in physically feasible and consistent reference signals.[55] Note, if $\dot{y}_{\text{ref}}$ was limited instead of $\dot{y}_{\text{des}}$,[56] then $\ddot{y}_{\text{ref}}$ could still change even if $\dot{y}_{\text{ref}}$ was stuck in a limitation - this behavior would be unphysical and the causality chain would be inconsistent.

In order to understand how the proposed second-order reference model from Fig. 4 allows to account for the inner-loop dynamics, it can be compared to the first-order reference model, depicted in Fig. 5. Since the angular rate is the physical entity commanded to the next control loop, it is commonly practiced that a first-order reference model generates the reference signals $\dot{y}_{\text{ref}}$ and y_{ref}. However, the commanded $\dot{y}_{\text{cmd}}$ cannot be reached instantaneously because of inner-loop dynamics.

For the following comparison, assume that these inner-loop dynamics are such that the true $\dot{y}$ follows $\dot{y}_{\text{cmd}}$ with first-order dynamics, as depicted in Fig. 5. Fig. 6 compares the step responses of the two systems from Fig. 5 and Fig. 4, denoted by subindex 1 and 2 respectively.

The second-order reference model from Fig. 4, along with the associated feedforward signal, establishes accurate tracking of the reference trajectory using only the feedforward signal, whereas the variant with first-order reference dynamics lags behind. The error controller then needs to contribute to the tracking. The second-order reference model, however, considers the inner-loop dynamics, similar to the reference models in Ref. 57. The chosen feed-forward signal equals $\dot{y}_{\text{des}}$ in the unsaturated case. If $K_2 = \bar{K}_2$ exact tracking can be achieved with the feed-forward only: We command $\dot{y}_{\text{des}}$ as feedforward but expect the "reference model inner-loop" as being a delay/-dynamics present in the plant/ true inner-loop, and therefore, we delay the reference dynamics by that amount. However, in order to also take into account the $\dot{y}$ saturations, we cannot directly take $\dot{y}_{\text{des}}$ as feedforward but "recalculate" an effective "saturated $\dot{y}_{\text{des}}$" through a scaled $\ddot{y}$ and $\dot{y}_{\text{ref}}$.

Because the feedforward achieves accurate tracking, the error controller can, hence, mainly counteract disturbances and model uncertainties. Note that choosing $\dot{y}_{\text{ref}}$ as feedfor-

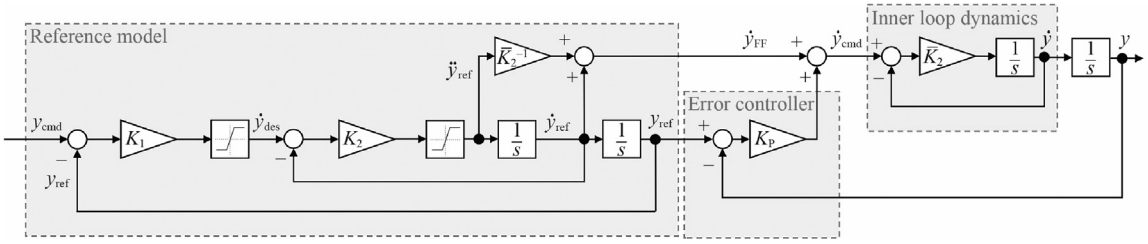


Fig. 4 Second-order model adapted from Ref. 41.

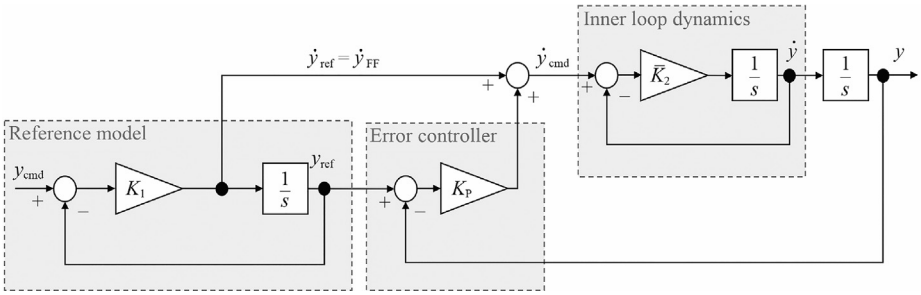


Fig. 5 Visualization of reference model, error controller, and inner-loop dynamics adapted from Ref. 41.

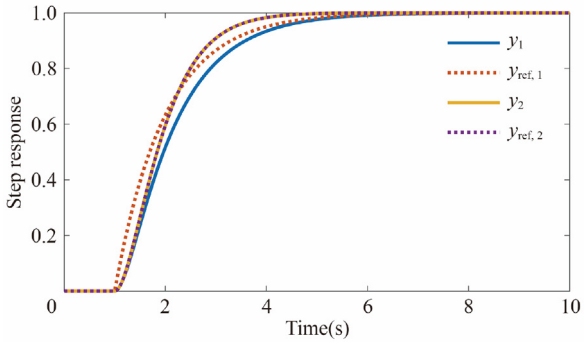


Fig. 6 Reference tracking without (y_1) and with (y_2) consideration of inner-loop dynamics in the reference model - no error controller, based on Ref. 41.

ward Ref. 58 leads to a delay in the feedforward phase and hence a delay in the inner-loop response.

4.3. Cascaded physical reference model

In 2018, Ref. 18 proposed an integrated reference model for a Tilt-rotor VTOL Transition UAV that consists of a decoupled outer and inner-loop reference model, which has been used in the context of Ref. 11. The outer-loop reference model provides load factors and velocities as reference states and consists of reference plant dynamics (point mass equations) in connection with a reference NDI controller. The NDI controller makes sure that the closed-loop reference system inherits desired dynamics. Instead of directly specifying desired dynamics by using, for example, a linear reference model, this approach allows the incorporation of protections and PCH as also described in Ref. 18. The inner-loop reference model calculates body angular accelerations and Euler angles and derivatives as reference trajectories. Ref. 18 also incorporated a command mapping and blending strategy.

4.4. Higher order reference model

In 2021, Ref. 45 presented a higher-order reference model that includes a physical design plant model. It generates command variable reference trajectories and the pseudo control derivative $\dot{\nu}_{FF}$, which serves as feedforward. We summarize the approach in the following. For simplicity, let us consider a single-state, SISO linear time-invariant system

$$\begin{cases} \dot{x} &= Ax + Bu \\ y &= x \end{cases} \tag{38}$$

and first-order actuator dynamics with bandwidth ω_a. The ANDI control law Eq. (23) in that case becomes

$$u_c = \frac{1}{B\omega_a}(\dot{\nu}_{des} - A\dot{x}) + u \tag{39}$$

where $F_u = B$, and $F_x\dot{x} = A\dot{x}$ was replaced and where

$$\dot{\nu}_{des} = \ddot{y}_{ref} + K_D(\dot{y}_{ref} - \dot{y}) + K_P(y_{ref} - y) \tag{40}$$

The reference system as proposed by Ref. 45 implements the system dynamics

$$\ddot{y}_{ref} \quad = \dot{x}_{ref} = A_R\dot{x}_{ref} + \dot{\nu}_{u,ref} \tag{41}$$

together with a NDI law in the reference model, given by

$$\dot{\nu}_{u,ref} = \dot{\nu}_{des,ref} - A_R\dot{x}_{ref} \tag{42}$$

Applied to the reference system Eq. (41), the closed-loop reference dynamics will result in $\ddot{y}_{ref} = \dot{\nu}_{des,ref}$, since the reference system, that is A_R, is exactly known. $\dot{\nu}_{des,ref}$ can, for example, result from linear desired reference dynamics, denoted by subindex R, combined with a reference model controller as

$$\dot{\nu}_{des,ref} = \dot{\nu}_{R,ref} + \dot{\nu}_{ref,ec} \tag{43}$$

The reference model structure with the reference system, reference model error controller, and linear reference model is depicted in Fig. 7.[45]

Note that in Fig. 7, $\dot{\nu}_x$ corresponds to $A_R\dot{x}_{ref}$. With this specific reference model design, we can approximate the ANDI control law Eq. (39), as was pointed out in Ref. 42, with

$$u_c = \frac{1}{B\omega_a}(\dot{\nu}_{des} - A_R\dot{x}_{ref}) + u \tag{44}$$

where $A\dot{x}$ was replaced by $A_R\dot{x}_{ref}$, and where

$$\dot{\nu}_{des} = \dot{\nu}_{des,ref} + \dot{\nu}_{ec} \tag{45}$$

The resulting controller is depicted in Fig. 8.[42] The resulting closed-loop dynamics are then

$$\begin{aligned} \ddot{y} &= A\dot{x} + B\omega_a(u_c - u) \\ &= A\dot{x} + B\omega_a\tfrac{1}{B\omega_a}(\dot{\nu}_{des} - A_R\dot{x}_{ref}) \\ &= A\dot{x} - A_R\dot{x}_{ref} + \dot{\nu}_{des} \end{aligned} \tag{46}$$

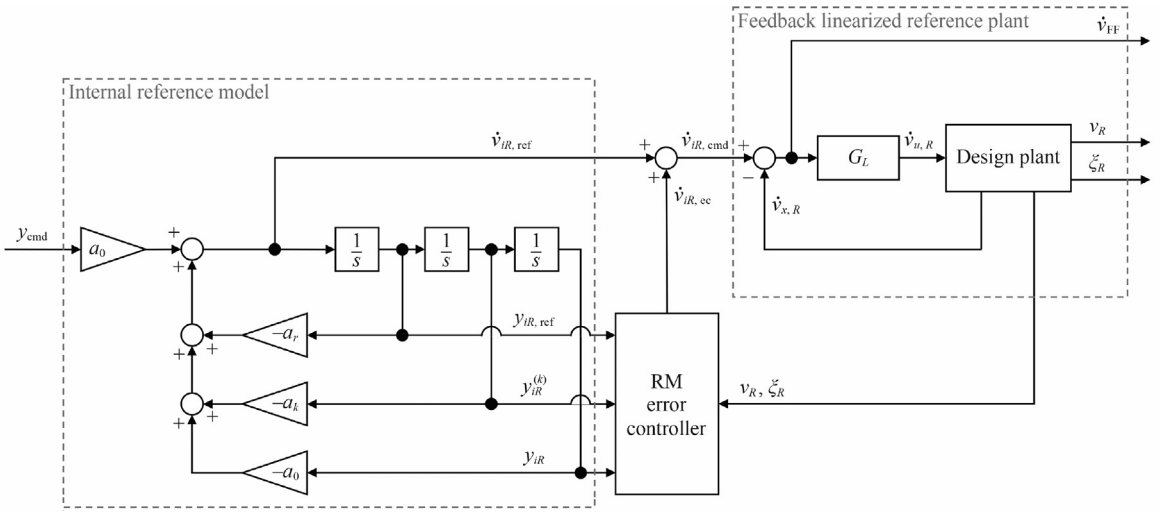


Fig. 7 Higher order reference model adapted from Ref. 45.

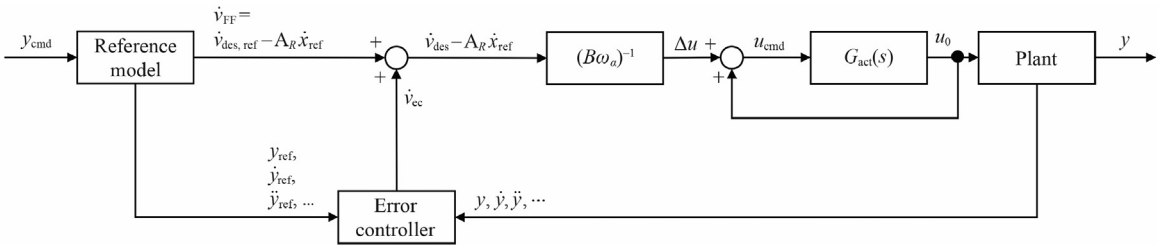


Fig. 8 ANDI controller overview adapted from Ref. 42.

In the ideal case, the state derivative-dependent terms cancel.

4.5. Incremental physical reference model

In 2022, Ref. 59 proposed a physical reference model for an INDI law to control the longitudinal motion of a piloted fixed-wing aircraft. Physical reference models have already been used and proposed in combination with NDI, for example, by Ref. 60, Refs. 43,61,62. The idea behind a physically motivated reference model is to provide reference signals with reasonable desired dynamics with respect to the actual aircraft dynamics. This ensures that the generated reference signals and their derivatives are consistent with the actual aircraft dynamics, allowing the response to be tracked in a physically realistic manner. It was found by Ref. 43, that a better emulation of the plant characteristics in the reference model reduces the control effort and tracking error in the transient. Furthermore, envelope protections can be intuitively embedded in the reference model. Fig. 9 depicts the physical reference model from Ref. 59. It consists of five key elements: a PI controller for the reference system, phase-plane-based protections, equivalent actuator or pseudo control dynamics, and an incremental short-period system model.

The incremental linear reference plant model is developed using the short-period approximation. Rather than using the traditional approach of defining the reference state based on the trimmed angle of attack, α_{trim}, the design instead employs the design maximum angle of attack. This method eliminates the need for trim tables, including α_{trim}, which may be uncertain or unknown. The pitch rate increment, however, remains defined relative to the trim pitch rate as is standard. Finally, it is necessary to link or transform the limit and trim domains.

In general, the transformation of the considered variables between the limit increment $\Delta x_{\text{lim}} = x - x_{\text{lim}}$ and the trim domain $\Delta x_{\text{trim}} = x - x_{\text{trim}}$ can be carried out according to

$$\Delta x_{\text{lim}} = \Delta x_{\text{trim}} - x_{\text{lim}} + x_{\text{trim}} \tag{47}$$

However, the trim value α_{trim} is not known. Likewise, the maximum pitch rate at the design maximum angle of attack is also uncertain. To address this, the load factor serves as the link between the incremental variables in both the trim and limit domains, where the trim and maximum reference values are known. This connection relies on the steady-state relationship between the load factor and pitch rate. Ref. 59 provided more details and the complete derivation.

The reference model controller is a classical PI controller with parameters designed to shape the closed-loop dynamics of the incremental short-period model to satisfy the handling quality requirements for the pilot commands. Note that the reference model controller design does not have to consider stability margins and robustness because the incremental short-period plant model is known precisely.

The pseudo control deficiency adjustment compensates for actuator position and rate limitations. It restricts the reference model to avoid generating reference trajectories that cannot be realized. Additionally, it ensures that the command will not

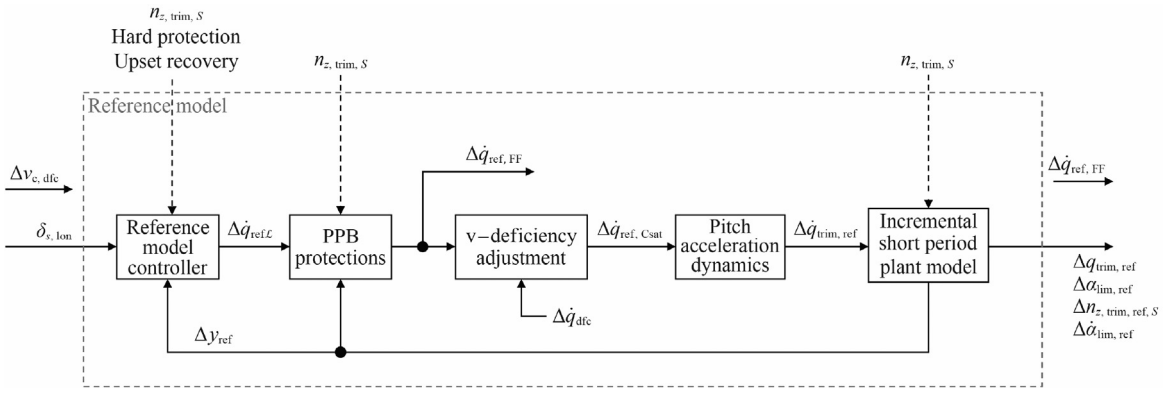


Fig. 9 Incremental physical reference model adapted from Ref. 59.

reverse as a result of saturation and will not be amplified without input from the pilot.

The linear dynamics, which specify the equivalent pseudo control dynamics, correspond to the estimated actuator dynamics.

Because of the reference model's physical character, control variable blending can be realized. Control variable blending provides intuitive control over the whole flight envelope. It allows the change of the steady-state control variable based on airspeed and stick deflection, for example. At high stick deflections, it might be, beneficial for the pilot to directly control the limiting flight envelope variable, that is, the angle of attack below the maneuver speed and load factor above the maneuver speed. At low stick deflections, a blend of pitch rate and load factor can provide intuitive behavior for the pilot. The error controller incorporates the blending. The physical reference model ensures that the variables to be blended are physically consistent, so the errors in the different variables do not fight against each other. Furthermore, the blending is accomplished by formulating the error controller with respect to the equivalent load factor. After performing the blending, the error controller's output is converted to a pitch rate acceleration for the INDI law.

4.6. Protections in reference models

In 2018, Ref. 18 described how PCH and simple envelope protections can be incorporated into a linear reference model or into an integrated physical reference model. In 2019, Ref. 63 applied phase plane-based protections in their reference model in combination with an INDI controller for a tilt-rotor VTOL transition UAV.

Phase plane-based protections have already been previously used and investigated [64–67] with other control laws. In 2019, Ref. 68 enhanced the phase plane-based Flight Envelope Protections (FEP) by an approach, they called Output limiting with Backstepping in the phase plane (short OLB FEP). Fig. 10 shows how it can be used as open-loop or closed-loop FEP in the reference model.

In 2019, Ref. 69 applied the concept of OLB envelope protection to a cascaded incremental flight controller for a commercial aircraft model. They applied an open-loop command limiting strategy within the outer-loop reference model. To account for disturbances, the inner-loop uses a closed-loop virtual control limiting pitch angle protection. Fig. 11 showed

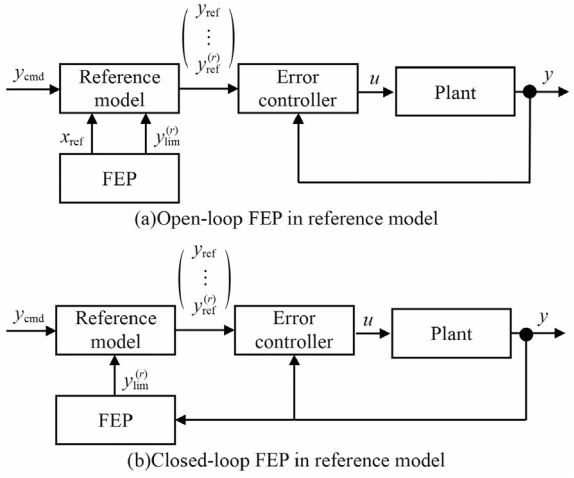


Fig. 10 Open-loop and closed-loop FEP in reference model adapted from Ref. 68.

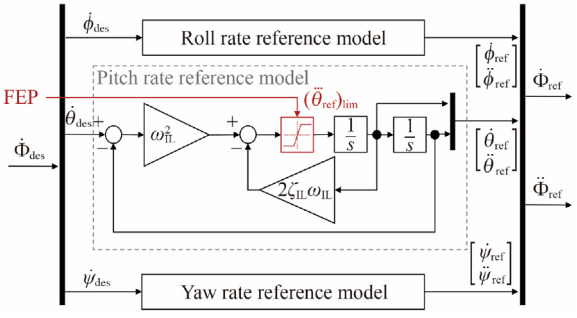


Fig. 11 Inner-loop reference model with flight envelope protection adapted from Ref. 69.

how to embed the protection in the inner-loop reference model.

In 2022, Ref. 70 thoroughly tested the FEP in the INDI law from Ref. 59, applying it to a piloted, all-attitude turboprop demonstrator aircraft. Ref. 70 used counter-optimization with Deep Q-Learning to test the envelope protections. This method evaluates the worst-case performance caused by the worst-case inputs and wind gust disturbances. It can be utilized

for complex closed-loop systems without requiring model simplifications. They demonstrated how to discover scenarios where the load factor protection is exceeded due to longitudinal stick inputs and vertical gusts.

4.7. Key takeaways on reference models and error controllers

This section summarizes key takeaways from Section 4 and discusses the relationship between the different reference model and error controller designs. Furthermore, this section provides an overview of their respective advantages and selection criteria.

The use of cascaded control loops where an outer control loop generates ν_{des}, see in Fig. 2, Section 4.1, is popular due to its low complexity. This approach provides good results. The controller gains shall be chosen to satisfy sufficient timescale separation between the control loops.

If specific requirements on the closed-loop dynamics must be satisfied, the controller gains must be carefully tuned. The process of tuning the controller gains at the same time with regard to performance, robustness, and stability requirements might be tedious.

By using a reference model in combination with an error controller this tasks can be decoupled. The reference model is parametrized to satisfy the requirements on the dynamics, and provides the desired dynamics. The error controller ensures that the reference dynamics are tracked. The tuning can focus on stability, robustness and disturbance rejection.

The utilization of linear reference models, in Section 4.2, is very popular due to their low complexity and because it is straightforward to define the desired dynamics. They have been applied in combination with various error controller structures, see in Section 4.2. A general statement about which structure is the best cannot be made - this is application dependent. A gain design process or routine may help to tune the controller gains, see in Section 10.

The linear reference models can be enhanced easily by minor structural adjustments, such that they provide physically feasible and achievable reference trajectories. Section 4.2, Fig. 4, describes the adjustments to the linear reference models in order to.

(1) account for absolute limits and rate and acceleration limits of physical entities.
(2) account for inner-loop dynamics.
(3) feedforward signals, to enable direct command feedthrough.

Instead of using linear reference dynamics, physical reference models can be used, often in combination with a reference model controller, see in Section 4.3 to Section 4.6. The key advantages and features for this choice are:

(1) Physical reference models provide reference signals with reasonable desired dynamics with respect to plant dynamics and physically consistent signal derivatives (allowing the response to be tracked in a physically realistic manner).
(2) They enable a decoupled design of desired closed-loop dynamics (via reference model and reference model controller) and stability, robustness and disturbance rejection (via error controller).
(3) Command mapping and blending strategies can be incorporated, as a physical reference model provides several variables or states that are physically consistent.
(4) Feedforward signals can be generated that include state-dependent terms which can be used in combination with the ANDI and E-INDI approaches.
(5) Protections and PCH can be embedded intuitively.

5. How to obtain control input

As derived in Section 3, the INDI control law needs measurements or estimates of the actual plant inputs u_0. However, those are not always directly measured or measured at such a low quality that we cannot use them as feedback signals. Here, we shortly summarize some of the most common ways that are embedded in INDI control laws to obtain the u_0 signal: (A) measurement of u_0, see Fig. 12 (a),[71] (B) time delayed actuator command, see Fig. 12 (b),[72] (C) based on actuator

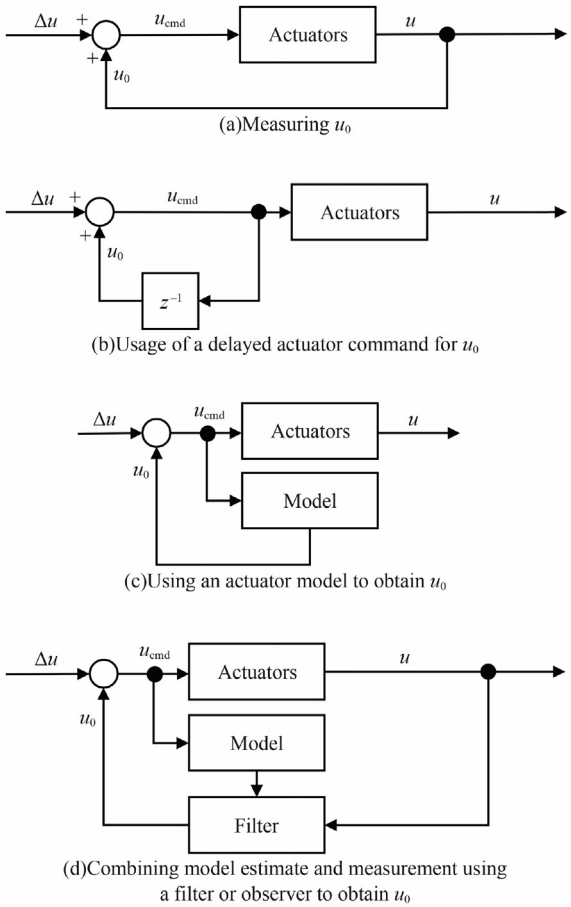


Fig. 12 Some common ways to obtain the u_0 signal.

model, see Fig. 12 (c),[11,47,59,73,74] (D) based on actuator model and measurement using Luenberger-like observers or advanced estimation algorithms and filters, see Fig. 12 (d)[75].

Feeding back the time-delayed command for very fast actuators offers a feasible solution because it reduces the required knowledge. This approach eliminates the need for additional sensors to measure the actuator output, avoiding the introduction of measurement noise or vibrations. It also removes the necessity for an actuator model and precise actuator parameter knowledge.

Ref. 74 addressed the special case where the actuator controller runs with a slower sampling rate than the flight controller. The controller accounts for this effect using a Zero-order Hold (ZoH) discretization of the actuator model.

In one of the initial papers on INDI, Ref. 76 recommended the usage of (first-order) actuator models even if measurements of the real actuators are available. The reason is that uncertainties in the measured u signal might accumulate due to the "integrating" behavior of the u feedback loop. By combining the first-order actuator model and the measurement in a smart way, this potential issue can be addressed.

The tracking of the control input command can sometimes be compromised by backlash and friction. This might cause problems in the outer control loops because the therein generated input commands are not tracked by the actuation or effectors.[77–79] A question for future investigation is how these effects might impact INDI-based control and what measures would be necessary to mitigate them.

So far we have discussed how to obtain the pseudo control command ν_{des}, as well as an estimate of the current actuator value u. What is missing in order to practically implement the INDI law, is an estimation of the current pseudo control value ν, which we address in the next section.

6. How to obtain pseudo control variable

As elaborated in the previous sections, pseudo controls and control inputs measurements in INDI replace a significant portion of the model-dependent part. Compared to NDI, this generally increases robustness against model uncertainties.

For a system with a relative degree of one, the pseudo control corresponds to the first derivative of the controlled output. For aircraft control applications, the pseudo controls are often angular accelerations and linear accelerations.[24,27,31,40,80] For both angular and linear accelerations, noise in the measurement signal might require advanced signal processing and filters.

Furthermore, although sensors for measuring angular acceleration exist, aircraft are often not equipped with these types of sensors for different reasons (e.g., costs and weight). In these cases, the pseudo control signal or output derivative has to be estimated.

Different ways of doing so exist and have evolved. As we will see in the following sections - one mainstream originated from the idea to obtain the angular acceleration via numerical differentiation of the angular rates; see in Section 6.1. This approach is not applicable in practice as the numerical differentiation amplifies measurement noise. One approach, addressed in Section 6.1, proposes to filter u_{cmd} to attenuate the noise on the command signal. However, high-frequency oscillations remain.

Instead of filtering u_{cmd} other approaches propose to low-pass filter the differentiated signal (resulting in a "high-pass" on the measurement of the rates - low-pass times s), see in Section 6.2. However, very soon, it was noticed that this will affect the response and stability due to the introduced phase lag of the low-pass filter, as will be detailed in Section 7.

An important insight was that under certain conditions, in case the angular accelerations can be perfectly measured, the closed-loop SISO response (from ν_{cmd} to ν) corresponds to the actuator dynamics. It was found, see in Section 7.1, that by applying the same low-pass on the u feedback as is used in the low-pass filtered differentiator, the transfer function from ν_{cmd} to ν will correspond again to the actuator dynamics (only the disturbance response remains low-pass filtered). This approach is often denoted as "synchronization" and Section 7 will address this topic in more detail. It will also show how to compensate for other delays that might be present in the system.

Besides this mainstream, further methods evolved to estimate the pseudo control signal, and we will illustrate them in Sections 6.6 to 6.8. In summary, we will elaborate on the following common approaches to estimating the pseudo control:

(1) Section 6.1: Numerical differentiator, optionally with an additional low-pass filter on u_{cmd} (e.g., for obtaining angular accelerations from angular rate measurements)
(2) Section 6.2: Numerical differentiator in combination with a low-pass filter on the pseudo control (e.g., for obtaining angular accelerations from angular rate measurements)
(3) Section 6.3: Linear mix of model-based and sensor-based pseudo control estimate (for pseudo control estimation in general, angular and linear accelerations)
(4) Sections 6.4 and 6.5: Frequency weighted mix of sensor-based and model-based pseudo control estimate via complementary filter (for pseudo control estimation in general, angular and linear accelerations)
(5) Section 6.6: Extended state observer (for pseudo control estimation in general, angular and linear accelerations)
(6) Section 6.7: Predictive filter (for pseudo control estimation in general, angular and linear accelerations)

6.1. Discrete derivative of the output measurement using a differentiator

Assume the pseudo control signals, e.g., angular accelerations, are not directly measured, but the angular rates are measured. In this case, one can apply pure numerical differentiation to the measured angular rates to estimate the pseudo control. Simplified, one subtracts the previous angular rate sample from the current angular rate sample and divides it by the sample time to approximate the angular acceleration.

However, pure numerical differentiation is not recommended due to its sensitivity to sensor noise, which can cause noise amplification in the control signal. In 2013, Ref. 81, estimated angular accelerations using this technique by applying backward finite differences to the rate measurement. They filtered u_{cmd} with a first-order low-pass filter to mitigate high-frequency oscillations resulting from noise introduced by numerical differentiation. Despite this, due to sensor dynamics

and measurement delays, low-amplitude, high-frequency oscillations persisted in the control commands. Additional examples of finite differences being utilized or explored can be found in Ref. 48 and Ref. 46.

6.2. Discrete derivative in combination with a low-pass filter

In 2001, Ref. 76 proposed using a discrete differentiator combined with a second-order low-pass filter, similar to Smith's approach. Smith also filtered and differentiated the pitch rate measurement q_0 to calculate $\dot{q}_0$ for his inversion law.[82] This method estimates the derivative for low-frequency signals while attenuating high-frequency sensor noise. This concept is widely used with different kinds of low-pass filters, e.g., first-order [49], or second-order.[22–24,30,47,54,83].

Low-pass filters introduce lag, leading to a delayed estimate of the relevant output derivative, degrading the controller's performance and stability. Several techniques were proposed to accommodate this, such as the previously mentioned "synchronization." Section 7 will address this topic.

6.3. Linear mix of model-based and sensor-based output derivative

The Hybrid INDI presented in 2020 by Ref. 15 uses in the INDI law a linear proportional blend between the measured acceleration $\dot{y}_0$ and the model-based $\dot{y}_{\mathrm{OBM}}$

$$\dot{\hat{y}} = K_{\mathrm{aug}}\dot{y}_0 + (1 - K_{\mathrm{aug}})\dot{y}_{\mathrm{OBM}} \tag{48}$$

where K_{aug} defines the percentage of the measured acceleration used. This allows for a specification of the balance between measurement and model-based estimation, which can be adjusted based on flight conditions. In flight regimes where the model is known to be accurate, K_{aug} can be decreased such that the model-based estimate is prioritized. The feedback reduction increases the classical stability margins in such cases where the model is accurately known, as Ref. 15 showed. In-flight conditions where the model is uncertain, for example, at a high angle of attack, the high model uncertainties can cause robustness issues. Then, the gain can be increased to emphasize the measurements and use more feedback to replace the uncertain model through the measurement. This reduces robustness issues due to uncertainties caused by the inaccurate model.

6.4. Complementary filter – Frequency weighted mix of sensor-based and model-based estimate

In 2005, we can find a reference on using a complementary filter in the context of incremental laws to calculate the rotational accelerations.[84] In Ref. 84, Dickman describes successful flight tests with Smith's incremental inversion law (from Ref. 33) on a 30 mm wingspan Micro Aerial Vehicle (MAV). However, they give no further details on how the complementary filtering is performed, except that it uses the measured rates.

In 2018, Ref. 31 proposed to estimate the angular acceleration using a complementary filter. The low-frequency part of the estimated output derivative $\dot{\hat{y}}$ is obtained from the low-pass filtered and differentiated measured angular rate y_0. We obtain the high-frequency part of the derivative estimate from the high-pass filtered model-based angular acceleration estimate $\dot{y}_{\mathrm{OBM}}$.

$$\dot{\hat{y}}(s) = (1 - L(s))\dot{y}_{\mathrm{OBM}}(s) + L(s)sy_0(s) \tag{49}$$

The filter $L(s)$ is a linear second-order low-pass filter such that the estimate results as

$$\dot{\hat{y}} = \frac{s^2 + 2\zeta\omega_0 s}{s^2 + 2\zeta\omega_0 + \omega_0^2}\dot{y}_{\mathrm{OBM}} + \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 + \omega_0^2}sy_0 \tag{50}$$

Fig. 13 shows in blue the frequency responses of the low-pass filter $L(s)$ and high-pass filter $1 - L(s)$ with $\zeta = 1$ and $\omega_0 = 20$ rad/s. We can see that the low-pass filter will counteract the amplification of measurement noise due to the differentiation as its magnitude decreases at high frequencies. The high-pass filtered model-based estimate will provide the high-frequency content of the final estimate, whereas the high-pass filter will remove any steady-state bias, because of its low magnitude at low frequencies. This kind of filter is used in several INDI controllers, for example, Refs. 31, 41, 85. calculated additionally the translational accelerations using a complementary filter by combining the low-pass filtered acceleration measurements with a high-pass filtered model-based acceleration estimate

$$\dot{\hat{y}}(s) = (1 - L(s))\dot{y}_{\mathrm{OBM}}(s) + L(s)\dot{y}_0(s) \tag{51}$$

The filter $L(s)$ is a linear first-order low-pass filter. The complementary filter attenuates the high-frequency noise of the measurements. The thereby removed high-frequency content of the signal is replaced by the high-frequency content provided by the model. With a higher bandwidth of the low-

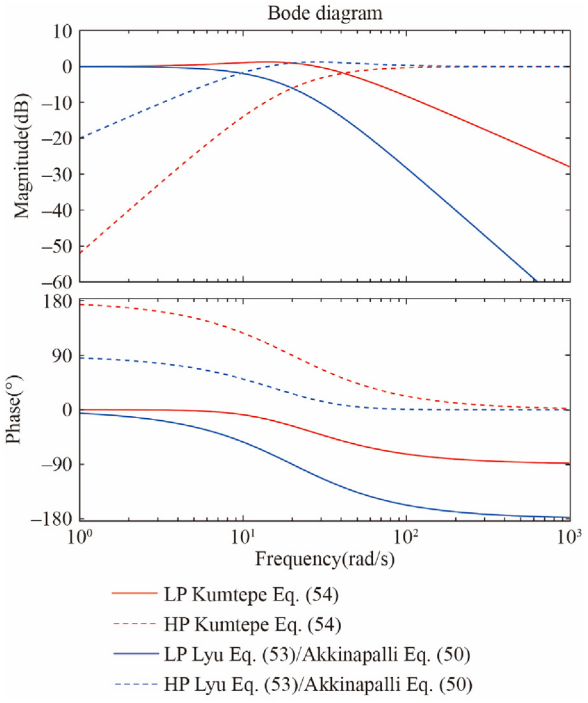


Fig. 13 Comparison of the frequency response of the proposed complementary filter in Eqs. (50), (53), (54).

pass filter, the acceleration estimate includes more measurement information. This means that a better response to disturbances and model uncertainties can be expected in the INDI law. However, an increased bandwidth will also reduce the noise attenuation. The complementary filter bandwidth could be, for example, determined by

(1) analyzing the noise frequency domain properties of the measurements and the estimation error, calculating the power spectral density.[86]
(2) adaptive approaches, for example, Least Square Estimation-Based.[87]
(3) holistic requirements-based multiobjective optimization for tuning both INDI controller gains and filter time constants.[88]
(4) gain scheduling-based on flight conditions.[89]

In 2022, Kumtepe et al. call the complementary filter approach Hybrid INDI[25] (this is not to be confused with Ref. 15, who call their static gain combination Hybrid INDI as well, compare with Section 6.3). For the Hybrid INDI, Ref. 25 propose a synchronization scheme on the actuator feedback in combination with a complementary filter given by

$$\dot{\hat{y}} = \frac{s^2}{s^2 + K_P s + K_I}\dot{y}_{OBM} + \frac{K_P s + K_I}{s^2 + K_P s + K_I} s y_0 \tag{52}$$

Note, that choosing $K_P = 2\zeta\omega_0$ and $K_I = \omega_0^2$, Eq. (52) results in

$$\dot{\hat{y}} = \frac{s^2}{s^2 + 2\zeta\omega_0 + \omega_0^2}\dot{y}_{OBM} + \frac{2\zeta\omega_0 s + \omega_0^2}{s^2 + 2\zeta\omega_0 + \omega_0^2} s y_0 \tag{53}$$

The complementary filters given in Eqs. (50), (53) differ in their choices of the numerator of the measurement and model part. Section 7 will present the synchronization of both variants. Ref. 25 compared Hybrid INDI and sensor-based INDI in a systematic manner based on an F-16 nonlinear simulation model. The results show that Hybrid INDI improves the tracking performance under nominal conditions (no uncertainties) in connection with the synchronization scheme. Simulations with uncertainties demonstrate that the Onboard Model (OBM) of Hybrid INDI does not need to be highly precise for the control system to remain robust against model discrepancies.

In 2023, Ref. 90 proposed a complementary filter with integrated synchronization, replacing the synchronization in the actuator feedback path. This approach is particularly advantageous when considering MIMO systems, where different filters and sensor dynamics in the feedback paths must be addressed or when input effectiveness is strongly coupled.[90] Section 7 addresses this in more detail.

6.5. Comparison of filtering methods

Ref. 90 provided an overview of some filtering-based methods to obtain derivatives and compare several synchronization techniques. They consider an INDI controller for a MIMO plant with different filters and sensor dynamics in the feedback paths. They analyze the effect of disturbances and the effect of state-dependent terms on the closed-loop, which result from the presence of actuator dynamics.

However, only in wing-borne flight are the state-dependent terms relevant. For multicopter/rotor systems with direct thrust, they are usually negligible. Based on the results of the comparison, Ref. 90 proposed that in case different filters and sensor dynamics in the feedback paths have to be accounted for, a complementary filter that incorporates the synchronization is preferred over applying the synchronization to the actuator feedback.

6.6. Extended state observer-based

In 2023, Ref. 91 proposed an Extended State Observer (ESO) to estimate angular accelerations based on an onboard mathematical model and rate measurements. The same approach is later used in Ref. 50. Lyu shows that the ESO can be expressed by

$$\dot{\hat{y}} = \frac{s^2 + l_1 s}{s^2 + l_1 s + l_2}\dot{y}_{OBM} + \frac{l_2}{s^2 + l_1 s + l_2} s y_0 \tag{54}$$

with the observer gains l_1 and l_2 determined using pole placement and by specifying the observer bandwidth. Comparison to the complementary filter Eq. (52), by choosing $K_I = l_2$ and $K_P = l_1$, shows that the two approaches mainly differ in their choice of the numerators of the model and measurement part. Ref. 91 showed empirically and by frequency domain analysis that ESO achieves superior noise attenuation compared to the complementary filter Eq. (52).

Fig. 13 shows the resulting frequency responses of the respective Low-pass filters (LP) given by $\frac{l_2}{s^2+l_1 s+l_2}$, Eq. (54) and $\frac{l_1 s+l_2}{s^2+l_1 s+l_2}$, Eq. (52), and the respective high-pass filters. The parameters are set to $l_2 = \omega_0^2 = (20\text{rad/s})^2$ and $l_1 = 2\zeta\omega_0$ with $\zeta = 1$. Lyu observed, as can be seen in the frequency responses, that the ESO Low-pass filter inherits a higher roll-off, resulting in higher noise attenuation of the differentiated measurement. Note that the ESO estimator is equal to the complementary filter in Eq. (50), as, for example, proposed by Refs. 31 and 90.

6.7. Predictive filter

In 2010, Ref. 20 found that the ideal INDI inversion law is susceptible to even slight time delays in the angular acceleration feedback and specifically address the difficulty in obtaining angular acceleration estimations required for the INDI inversion laws. They design a linear predictive filter to estimate the angular rate derivatives from angular rates and references thereof. They compute the coefficients of the predictive filter using least squares estimation. The inversion law, including the predictive filter, is denoted as "PINDI," Predictive Incremental Nonlinear Dynamic Inversion.[20]

6.8. Other methods

Ref. 9 utilized a setup of accelerometers that, along with a measured rotational rate, which is differentiated, is used to calculate the current rotational accelerations required by the inversion law using a least squares solution. Based on the angular acceleration, the measured linear accelerations are translated back to the Center of Gravity (CG) and are averaged. Later, however, they derive the angular acceleration from rate measurements using a second-order differentiator fil-

ter see in Section 6.2, as the setup turned out to be complex and sensitive towards structural vibrations.[76]

Ref. 92 proposed a Gaussian process regression method as a state derivative estimation technique and apply it to an incremental controller. They compare the method to the backward difference differentiator, a first- and second-order low-pass filter, and a high-order sliding mode differentiator. Ref. 93 proposed an adaptive method to obtain the derivatives needed for INDI. Some publications propose the usage of Kalman filters in combination with incremental laws such as Refs. 94–96.

7. Synchronization

In practical applications, control systems involve different kinds of delays, such as control signal transmission delays (actuator command delay), sensor signal delays, or processing and scheduling delays. Additionally, signal filtering introduces phase lag to the system.

The use of signal filtering in control laws is often inevitable and necessary due to reasons such as roll-off filtering for noise attenuation, notch filtering for attenuation of structural mode excitation and anti-aliasing filtering. Furthermore, as discussed in Section 6, pseudo control signals like angular accelerations may not be measurable and must be obtained via different filtering techniques. These filtering techniques may introduce additional phase lag in the feedback channel.

Already Ref. 33 in 2000 remarked that the pitch rate and pitch acceleration paths must be carefully matched." In 2001, Ref. 76 evaluated the control system's stability. They encountered low gain and phase margins in the normal accelerometer, which they used to derive angular acceleration data as described in Section 6.8.

In 2010, Ref. 20 found that the ideal INDI inversion law is susceptible to even slight time delays in the angular acceleration feedback as it is, for example, introduced when a differentiation filter in Section 6.2 is used. They proposed a method to deal with these measurement delays using predictive filtering to obtain the required angular acceleration, see in Section 6.7.

In 2013, Ref. 81 applied INDI to a helicopter. The investigations performed there showed that regarding the existence of a time delay between the control system and the actuators, the behavior of the helicopter remained acceptable, with lags of up to 50 ms.

Also Ref. 83, observed in 2018 that INDI is more sensitive to output derivative (pseudo control) measurement delay than actuator delay. They showed via simulation that unsynchronized delays between the actuator and angular rate feedbacks lead to oscillations. Furthermore, they systematically investigated the analytical stability of an exemplary single-input, single-output first-order linear system with a first-order actuator and an INDI controller. They obtained the state derivative via discrete differentiation and calculated the stability margins for different input/output delay values. The findings from these studies indicate that the system where both the actuator and state derivative measurements experience equal delays exhibits the largest stability region, compared to systems where delays affect only the actuator or state derivative measurements.

From then on, it became clear that the actuator and pseudo control signal feedback (often angular acceleration) must be carefully synchronized in time.

The research conducted in this field can be divided into synchronization schemes or measures to account for phase delay, sensor dynamics, and time delays, which the following sections will detail:

(1) Section 7.1: Delays in the pseudo control signal (acceleration) measurement/estimate.
(2) Section 7.2: Delays in the actuator command path.

If we consider the corresponding publications, the proposed schemes usually have to be considered in combination with how the derivative signal (acceleration estimate) is obtained. We saw in Section 6 that various ways to derive the derivative evolved, and each variant requires its suitable synchronization measure.

7.1. Accounting for delays in pseudo control signal path

In 2015, Ref. 22 introduced a method to address delays that arise when calculating angular accelerations from angular rate measurements using a differentiation filter (see in Section 6.2), providing further insights into the INDI closed-loop behavior. As the angular rate measurement is noisy, numerical differentiation amplifies the noise so that additional filtering is required. This filtering introduces a delay in the measurement, which should be compensated. They show that adding a synchronization filter on the measured actuator feedback u_0 is important, which equals the filter applied to the acceleration measurement as depicted in Fig. 14. Note, that $sH(s)$ is the differentiation filter, $F_{\text{cy}}(s)$ represents sensor dynamics and $\overline{H}(s)\overline{F}_{\text{cy}}(s) = H(s)F_{\text{cy}}(s)$ for the SISO case. Furthermore, $G_A(s)$ denotes the actuator dynamics, $G(s)$ is the plant dynamics, and d is a disturbance input. The closed-loop transfer function from the commanded pseudo control v_{des} to the pseudo control response equals then the actuator dynamics. Ref. 22 derived this closed-loop transfer function analytically in discrete time.

Similar to Ref. 22, Ref. 83 proposed to synchronize the actuator feedback path for the delays in the acceleration feedback path by adding the relative delay and using a synchronization filter that corresponds to the second-order filter that they used to filter the differentiated rates.

In 2018, Ref. 97 stressed that "the key for practical implementation is in ensuring synchronization between angular acceleration and control deflection measurements or estimates." They acquire the estimated angular accelerations by differentiating the measured rates and filtering with a second-order low-pass filter. They choose the filter such that

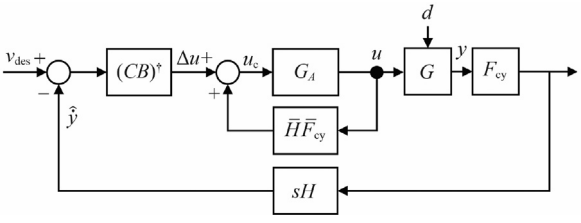


Fig. 14 INDI with derivative filter and synchronization by Ref. 22, figure adapted from Ref. 90.

the acceleration feedback is sufficiently free of noise, avoiding noise propagation into the control laws, as noisy control signals can lead to excessive wear on the actuators. The actuator feedback and acceleration feedback signals are synchronized by adding the relative delay between the two signals to the signal with the lower delay, such that they are delayed equally. As proposed by Ref. 22, the second-order filter used for filtering the rate derivative is added as a synchronization filter to the actuator feedback. They observe that in accordance to Ref. 83 and Ref. 81, INDI seems to be more sensitive to a surplus of state derivative delay compared to a surplus of input delay.

Also in later publications, e.g.,[24,47,54] the synchronization technique, as used by Ref. 22, is applied whenever a low-pass differentiation filter is used in combination with feedback of the control input u_0, measured after the actuator dynamics or at actuator model output.

In 2021, Ref. 23 proposed applying an input scaling gain Λ to increase the controller robustness to delays in the feedback loop when using the second-order filter differentiator. A drawback of the approach is the reduced bandwidth of the closed-loop response. The INDI control law given in Eq. (16) becomes in this case

$$u = u_0 + B_v^{-1}(x_0, u_0)\Lambda(v_{\text{des}} - \dot{y}_0) \tag{55}$$

They evaluate the concept in the time and frequency domain-based on an SISO system, where the plant dynamics are an integrator. Additionally the approach is tested for the lateral motion control of an airship. The nonlinear simulation model includes the vehicle's aerodynamics, actuators, and sensors.

In addition to the second-order filter differentiator, complementary filters have been suggested for estimating angular accelerations (see in Section 6.4).

In 2023, Ref. 90 reviewed different synchronization schemes analytically and via simulations. The analysis is based on a MIMO linear time-invariant system, with a relative degree of one, given by

$$\begin{cases} \dot{\boldsymbol{x}}(t) = \boldsymbol{A}\boldsymbol{x}(t) + \boldsymbol{B}\boldsymbol{u}(t) + \boldsymbol{B}_d\boldsymbol{d}(t) \\ \boldsymbol{y}(t) = \boldsymbol{C}\boldsymbol{x}(t) \\ \boldsymbol{v}(t) = \dot{\boldsymbol{y}}(t) = \boldsymbol{C}\dot{\boldsymbol{x}}(t) = \boldsymbol{C}\boldsymbol{A}\boldsymbol{x}(t) + \boldsymbol{C}\boldsymbol{B}\boldsymbol{u}(t) + \boldsymbol{C}\boldsymbol{B}_d\boldsymbol{d}(t) \end{cases} \tag{56}$$

They derive and compare the analytical closed-loop relations for

(1) INDI with derivative filter as described in Section 6.2 and synchronization on the actuator feedback path, as depicted in Fig. 14, and used in Refs. 22,83,97.
(2) Complementary filter
Eq. (53) with synchronization on the actuator feedback path, as depicted in Fig. 15, used in Ref. 25.
(3) Complementary filter Eq. (50) with incorporated synchronization, as depicted in Fig. 16, used in Ref. 90.

Note that $\boldsymbol{I}$ represents the identity matrix, and $\boldsymbol{F}_{\text{cx}}(s)$ represents sensor dynamics. Ref. 90 also considered Vanilla INDI (ideal INDI with perfectly known output derivative/pseudo control) and INDI with derivative filter but without synchronization. In their comparisons, they consider the closed-loop pseudo control tracking and the influence of the states and disturbances on the analytical closed-loop response in the pseudo control measurement for the MIMO case. In case that the measurement signals experience different delays and filters,

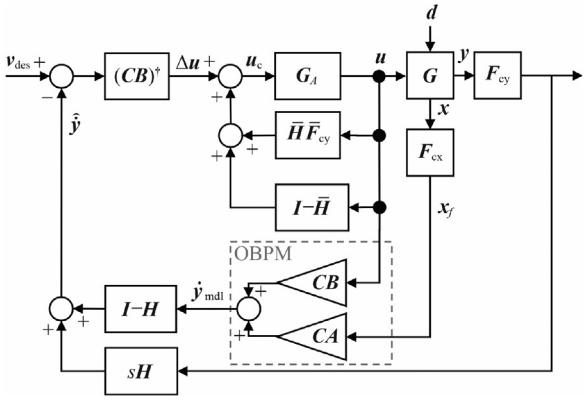


Fig. 15 Hybrid INDI with complementary filter and synchronization as proposed in Ref. 25 and plotted in Ref. 90.

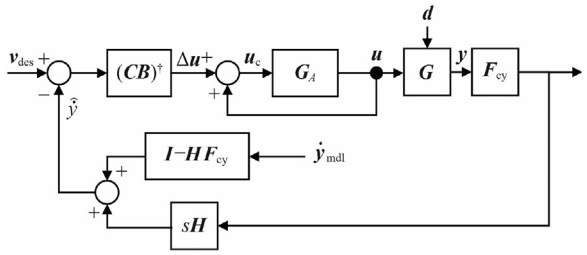


Fig. 16 INDI with complementary filter and incorporated synchronization adapted from Ref. 90.

the proposed approach with incorporated synchronization in the complementary filter, depicted in Fig. 16 is beneficial.

Ref. 90 performed simulations using the linearized lateral dynamics of an aircraft. In case different filters are applied to the different measurement signals, and additionally, the input effectiveness ($\boldsymbol{B}$ matrix) is strongly coupled, the synchronization on the actuator feedback may become unstable. In contrast, the integrated complementary filter approach recovers the design closed-loop tracking behavior of Vanilla INDI. For conventional fixed-wing aircraft, the $\boldsymbol{B}$ matrix is usually not strongly coupled, and the synchronization on the actuator feedback works reasonably well. This synchronization approach might be the preferred choice as it does not require additional model information. But, for example, in a V-tail aircraft the control inputs are coupled in the yaw and roll channel so that the complementary filter-based synchronization approach may lead to better results.

Ref. 90 gained further insights by inspecting the resulting closed-loop responses, which assume that the actuator transfer function $\boldsymbol{G}_A$ is diagonal and all elements are the same (equal actuator dynamics):

Assuming accurate model knowledge, that is $\dot{\boldsymbol{y}}_{\text{mdl}} = v$, they obtain for the complementary filter with incorporated synchronization, Fig. 16, the following closed-loop dynamics

$$s\boldsymbol{Y} = \left(\boldsymbol{I} - \bar{\boldsymbol{G}}_A\right)\boldsymbol{C}\boldsymbol{A}\boldsymbol{X} + \bar{\boldsymbol{G}}_A\boldsymbol{V}_{\text{des}} + \left(\boldsymbol{I} - \bar{\boldsymbol{G}}_A\boldsymbol{H}\boldsymbol{F}_{\text{cy}}\right)\boldsymbol{C}\boldsymbol{B}_d\boldsymbol{D} \tag{57}$$

which equals Vanilla INDI, except for the disturbance response which is in Vanilla INDI equal to $\left(\boldsymbol{I} - \bar{\boldsymbol{G}}_A\right)$. Assuming that all actuators have equal dynamics, then $\boldsymbol{G}_A$ is a diag-

onal square matrix where all diagonal entries are the same, namely the actuator dynamics. The dimension of $\boldsymbol{G}_A$ equals the number of inputs. $\bar{\boldsymbol{G}}_A$ is a diagonal matrix with the actuator dynamics on its diagonal elements but with different dimensions compared to $\boldsymbol{G}_A$. The dimension of $\bar{\boldsymbol{G}}_A$ equals the number of outputs.

For INDI with derivative filter and synchronization on the actuator feedback path

$$s\boldsymbol{Y} = \left(\boldsymbol{I} - \bar{\boldsymbol{G}}_A \boldsymbol{H} \boldsymbol{F}_{\text{cy}}\right) \boldsymbol{C} \boldsymbol{A} \boldsymbol{X} + \bar{\boldsymbol{G}}_A \boldsymbol{V}_{\text{des}} + \left(\boldsymbol{I} - \bar{\boldsymbol{G}}_A \boldsymbol{H} \boldsymbol{F}_{\text{cy}}\right) \boldsymbol{C} \boldsymbol{B}_d \boldsymbol{D} \tag{58}$$

is derived, where it had to be additionally assumed that $\boldsymbol{H}$ and $\boldsymbol{F}_{\text{cy}}$ are diagonal with the same elements, that is, all measurements are filtered equally. For the INDI with complementary filter and synchronization on the actuator feedback path,

$$s\boldsymbol{Y} = \left(\boldsymbol{I} - \bar{\boldsymbol{G}}_A \boldsymbol{F}_{\text{cy}}\right) \boldsymbol{C} \boldsymbol{A} \boldsymbol{X} + \bar{\boldsymbol{G}}_A \boldsymbol{V}_{\text{des}} + \left(\boldsymbol{I} - \bar{\boldsymbol{G}}_A \boldsymbol{H} \boldsymbol{F}_{\text{cy}}\right) \boldsymbol{C} \boldsymbol{B}_d \boldsymbol{D} \tag{59}$$

is derived under the same additional assumption that all sensors and filters are the same - an assumption that was not necessary for the approach with the complementary filter with integrated synchronization from Fig. 16. Note that the pseudo control tracking and disturbance response are the same for all approaches. They only differ in the disturbance caused by the high-pass filtered states, where the crossover frequency is the lowest for the INDI with derivative filter (slower changing states still have an influence) and the highest for the complementary filter with incorporated synchronization. As mentioned, the latter can also handle cases where the measurements are not equally filtered. For example, this might be necessary to attenuate structural modes in different channels. In Ref. 90, also a version or an extension of the other complementary filter approach has been developed that allows for different filters in the different feedback channels.

Ref. 32 introduced a synchronization for their Hybrid INDI from 2020,[15] where they use a proportional linear blend between model-based and sensor-based derivative, see in Section 6.3. They consider several filters in the angular acceleration feedback path as depicted in Fig. 17. The time delay of the angular acceleration measurement caused by the differentiation filter, sensor dynamics, and SCF (second-order structural coupling filter) causes asynchronicity between the measured acceleration feedback and the control surface feedback. Therefore, the stability margin of the control system is reduced.

They design a synchronization filter for the actuator feedback path to increase the stability margins. Instead of synchronizing exactly, they perform a Low Order Equivalent System (LOES) matching to obtain a second-order synchronization filter. The phase frequency response of the LOES synchronization filter they thereby match with the exact high-order angular acceleration feedback path, including all filters. Ref. 16 investigated different synchronization filters:

(1) 4th-order synchronization filter on u_{cmd} feedback path (before actuator).
(2) 2nd-order synchronization filter on u_0 feedback (after actuator).

Both are optimized in frequency to match the high-order acceleration feedback path. Furthermore, they investigate both synchronization filters, but this time with parameters optimized for margins. The second approach gives better results with regard to the aim to maximize the margins. They emulate structural coupling tests with all variants and an open-loop linear control system environment. They include the structural vibration signals on pitch rate, normal acceleration, and input commands obtained from ground tests. They conclude that to prevent structural vibration of the control surfaces from entering the control law, u_{cmd} shall be fed back. This way, the feedback of structural pickup and loads acting on surfaces is avoided.

Ref. 91 proposed an extended state observer-based angular acceleration estimation, together with synchronization, which Ref. 50 also used. The structure is depicted for a scalar SISO system in Fig. 18.

Ref. 91 validated the presented controller and the ESO and synchronization scheme in flight tests on a large-slenderness-ratio supersonic aircraft. The synchronization filter is given, based on Eq. (54) by

$$A(s) \cdot \left[\frac{l_2}{s^2 + l_1 s + l_2} S(s) + \frac{s^2 + l_1 s}{s^2 + l_1 s + l_2}\right] \tag{60}$$

where $S(s)$ is the sensor model and $A(s)$ the actuator dynamics. Though the estimator equals in structure the complementary filter approach from Refs. 31 and 90, as was mentioned in Section 6.6, the synchronization strategy differs from that in Ref. 90, which Fig. 16 depicts. The interested reader may find details regarding the synchronization in Ref. 91.

7.2. Accounting for delays in actuator command path

Ref. 98 investigated for the closed-loop system with first-order actuators, depicted in Fig. 19 how an over underestimation of:

(1) Control effectivity $\frac{\widehat{M_u}}{M_u}$.
(2) Actuator time constant $\frac{\widehat{T}}{T}$.
(3) Time delay $\frac{\hat{\tau}}{\tau}$.

affects the closed-loop pole locations.

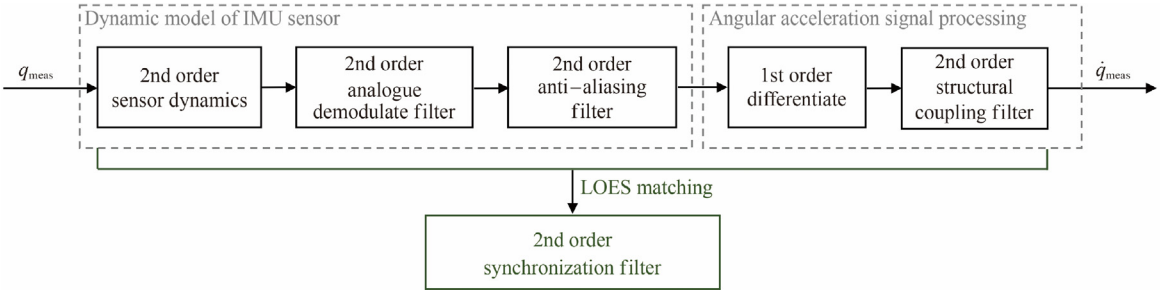


Fig. 17 Angular acceleration feedback path adapted from Ref. 32.

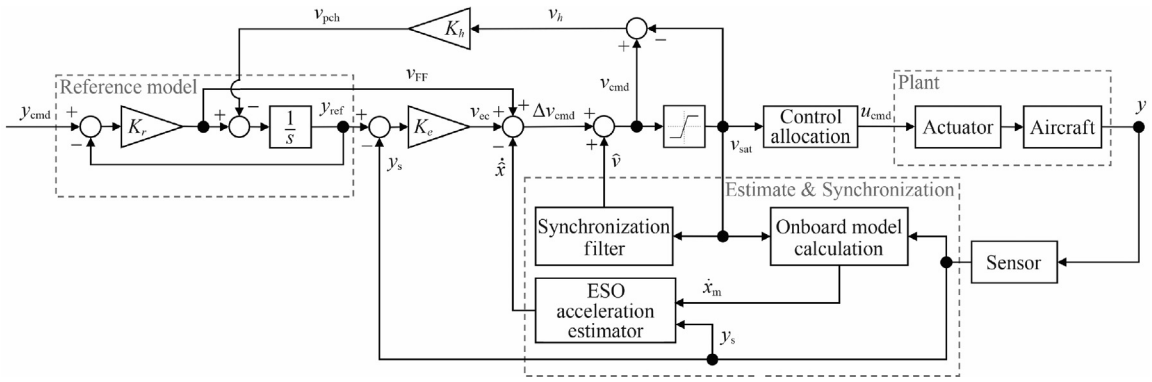


Fig. 18 Controller adapted from Ref. 91 with extended state observer-based angular acceleration estimation and synchronization.

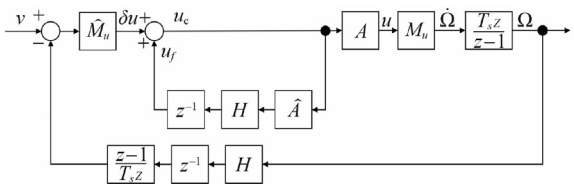


Fig. 19 INDI controller with actuator model adapted from Ref. 98.

Ref. 99 addressed delays and a linear second-order filter in the actuator command path in INDI control laws as depicted in Fig. 20.

The delays can originate from transmission or processing, as the digital flight control computer transmits the commands to the actuator control units. The post-processing filter might smooth the control command or prevent structural modes from being excited. Ref. 99 showed analytically that the steady-state error between the commanded actuator rate and the actual rate is zero in the classical INDI case (no delay and no post filter). As soon as a delay or a post filter is present, they show that the steady-state error becomes non-zero. The proposed approach of Ref. 99 in order to obtain zero steady-state error in the actuator rates $\dot{u}_{\mathrm{cmd,ss}} - \dot{u}_{\mathrm{ss}}$ is depicted in Fig. 21: Compared to conventional INDI, they pass the positive feedback of the actuator position through a special transfer function G_u in order to eliminate the steady-state error in the actuator rate. In addition, inspired by Ref. 40, they use actuator model parameters given by K_{act} and G_{act}, and desired parameters K_{des} and G_{des} to replace the actuator dynamics if the current actuator dynamic characteristics are not satisfactory. Therein $G_{\mathrm{des}} = \frac{2\zeta_{\mathrm{des}}\omega_{\mathrm{des}}}{s+2\zeta_{\mathrm{des}}\omega_{\mathrm{des}}}$ and $K_{\mathrm{des}} = \frac{\omega_{\mathrm{des}}}{2\zeta_{\mathrm{des}}}$, where ω_{des} and ζ_{des} are the desired actuator parameters.

Ref. 100 proposed an INDI-based actuator delay compensation control scheme, using a finite integral compensation based on model reduction.

7.3. Key takeaways on pseudo control estimation and synchronization

This section summarizes key takeaways from Sections 6, 7.1, 7.2 and provides several recommendations with regard to the estimation of the pseudo control and the related synchronization. In many cases, it is sufficient to use a low-pass filter (Section 6.2) and and synchronize with the same low-pass filter on the actuator feedback (Section 7.1, Fig. 14). Due to the low complexity of this approach, and because it does not require additional model information it is especially recommended if the following conditions hold true: (A) The system is SISO, (B) The state dynamics are much slower than the actuator dynamics ($\nu \approx B_\nu \cdot u$).

If the system is MIMO, this approach also works well in case that (A) $\boldsymbol{B}_\nu$ is approximately diagonal, (then the inputs are not strongly coupled - this is often the case for conventional aircraft), (B) the delays, sensor dynamics and filters on all feedback signals that are used to compose the pseudo control vector are equal.

Note that this approach is equal to a synchronized complementary filter in case that $\nu = \boldsymbol{B}_\nu \cdot \boldsymbol{u}$, meaning that there are no state-dependent terms $\boldsymbol{a}_\nu(\boldsymbol{x})$. Usually, these state-dependent dynamics, if they exist, are at lower frequencies than the dynamics related to $\boldsymbol{u}$. If in this case, the noise amplitude on the measurements is high, or if the noise frequency is low, the low-pass filter would need to have a low bandwidth in order to achieve a feasible noise attenuation. This might lead in addition to an attenuation of the state-dependent part, that is inherently contained in the unfiltered measurement signal. In that case, the hybrid INDI approach with complementary filter

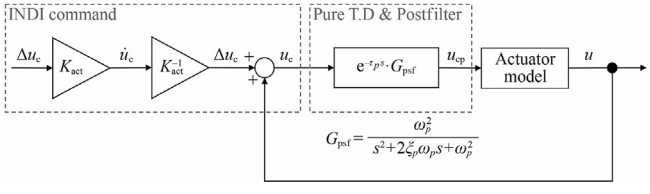


Fig. 20 Considered delays and filter in actuator command path adapted from Ref. 99.

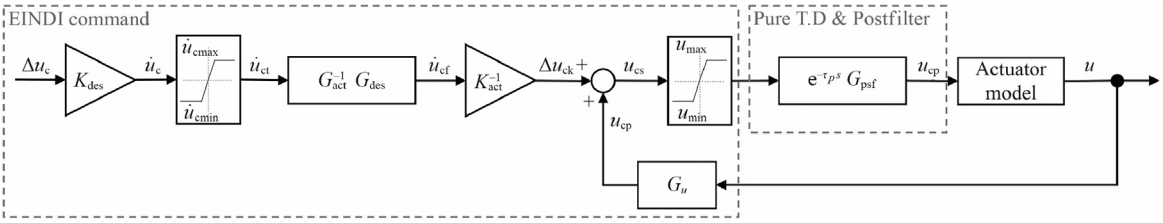


Fig. 21 Updated command structure to obtain steady-state accuracy in actuator rates adapted from Ref. 99.

from Section 6.4 is recommended. The hybrid INDI restores the high-frequency content via a model-based estimate (that contains the state-dependent part), combined with the low-pass filter on the measurement. The complementary filter in Eq. (50) corresponds to the Extended State observer in Section 6.6 and is recommended as it attenuates noise slightly better than the estimate from Eq. (53), see in Section 6.6.

For the hybrid INDI with complementary filter the complementary - filter based synchronization approach, Fig. 16, may be used for synchronization as it can deal with: (A) $\boldsymbol{B}_v$ that is not diagonal (for a V-tail aircraft, for example, the control inputs can be heavily coupled in the roll and yaw channel, but also for other configurations); (B) cases where the measurements are not equally filtered or delayed (this might be necessary, for example, to attenuate structural modes in different channels).

So far, we have discussed obtaining the desired pseudo control command, the control input measurement/estimation, and multiple ways of estimating the current pseudo control, where careful synchronization is required. Both, the desired pseudo control and current pseudo control estimate/measurement form the desired pseudo control increment, from which a required change in the control input must be calculated. The part responsible for this calculation is the "control allocation", as detailed in the next section.

8. Control allocation

Control allocation is a crucial component in NDI and subsequentially INDI control of overacuated systems. In the scope of this work, we define an overactuated system as a system with more inputs n_u than pseudo controls n_v.[38] Generally, control allocation is a broad and complex field with a large body of literature. As this paper focuses on INDI, we only address the specific aspects of control allocation relevant for INDI control. For a more detailed review of control allocation methods, refer to Refs. 38,101.

In non-incremental control, linear control allocation solves the underdetermined equation system

$$\boldsymbol{B}_v\boldsymbol{u} = \boldsymbol{v}_{\text{des}} \tag{61}$$

for $\boldsymbol{u}$, where $\boldsymbol{B}_v \in \mathbb{R}^{n_v \times n_u}$.[38] This problem can also be viewed as a linear optimal control problem as shown by Ref. 102, who compared optimal control with control allocation.

In contrast to linear optimal control, like LQR for example, control allocation takes the limitations of the actuation system into account, which usually consists of absolute limits $\underline{u}_i \leqslant u_i \leqslant \overline{u}_i$ and rate limits $\underline{\dot{u}}_i \leqslant \dot{u}_i \leqslant \overline{\dot{u}}_i$. Further limits like acceleration and jerk limits can also be accounted for Ref. 41. The limits result in a convex box admissible control set, which we define in the sense of Ref. 38 as the attainable moment set (AMS). This is then called constrained control allocation.

A desired pseudo control vector outside the AMS is called infeasible.[103] Thus, no exact solution exists, and an error metric could be minimized instead. The control allocation may find infinite solutions for feasible desired pseudo control vectors within the AMS.[103] To find a solution, a secondary objective like minimum energy consumption or utilization of preferred actuators can be incorporated.[101]

Commonly, a quadratic cost function will be minimized, and the following optimization problem can be formalized.

$$\begin{aligned} &\min_{\boldsymbol{u}} J(\boldsymbol{u}) = \frac{1}{2}\boldsymbol{u}^{\mathrm{T}}\boldsymbol{W}\boldsymbol{u} \\ &\text{s.t. } \boldsymbol{B}_v\boldsymbol{u} = \boldsymbol{v}_{\text{des}} \\ &\quad \underline{u}_i \leqslant u_i \leqslant \overline{u}_i \end{aligned} \tag{62}$$

The unconstrained optimization problem can be solved using a single pseudo-inverse. Quadratic programming solvers like the active set method can be used to consider the constraints.[101,104] Unfortunately, quadratic programming has a high or even unbounded computational complexity, such as through cycling effects. Therefore, an application to safety–critical systems like aircraft is a challenge.

Alternative suboptimal solution methods are the Redistributed Pseudoinverse (RPI) and the Redistributed Scaled Pseudoinverse (RSPI) from the group of cascaded generalized inverses.[17,38,105,106] The core concept of redistributed pseudo-inverse control allocation is applying the pseudo-inverse of the input matrix to the desired pseudo control vector. Then, the resulting input vector elements are clipped or scaled to limits, and the saturated effectors are omitted in the further allocation within the current controller time step. The residual (or unachieved) pseudo control vector is calculated, and the next cycle is started.[17] This process is continued until the residual is zero or the maximum steps have been reached. The resulting algorithm has a bounded execution time.

Ref. 17 described a RSPI implementation, which has successfully been flown onboard of an eVTOL transition drone. This implementation has been extended in Ref. 107 to incorporate SVD-based protections. Though the algorithm has been demonstrated to perform well for the control allocation of an overactuated and nonlinear aircraft, the online calculation of the SVD of the control effectiveness matrix may be challenging in the context of safety–critical software development, where the algorithms and their runtime have to be deterministic at all times. Since, in the considered applications, the control effectiveness matrix may show a high nonlinear dependence on states, a repeated online inversion of some kind may be inevitable. In order to address this challenge, Ref. 108

presented a numerically efficient, deterministic, and transparent rank-one update-based algorithm for the online calculation of inverses in the context of RSPI, with a special focus on safety–criticality.

So far, we only considered control allocation on the absolute commands, but in INDI, the control allocation takes place on an incremental level. Applying a redistributed pseudo-inverse method provides an approximate solution to the following optimization problem.

$$\begin{aligned} &\min_{\Delta \boldsymbol{u}} J(\Delta \boldsymbol{u}) = \tfrac{1}{2}\Delta \boldsymbol{u}^{\mathrm{T}} \boldsymbol{W} \Delta \boldsymbol{u} \\ &\text{s.t. } \boldsymbol{B}_v \Delta \boldsymbol{u} = \Delta \boldsymbol{v}_{\mathrm{des}} \\ &\quad \underline{\Delta u_i} \leqslant \Delta u_i \leqslant \overline{\Delta u_i} \end{aligned} \tag{63}$$

It must be noted that the redistributed pseudo-inverse method does not necessarily give the global optimum.

Since we are now minimizing the input increments $\Delta \boldsymbol{u}$ and not absolute inputs $\boldsymbol{u}$, the so-called path dependency problem arises.[38,109] This problem emerges when the system is overactuated, characterized by a broad input matrix $\boldsymbol{B}$ with null space $\aleph_B$. Due to the null space of $\boldsymbol{B}$, zero pseudo control vector $\boldsymbol{v}_{\mathrm{des}} = \boldsymbol{0}$ does not imply a zero control input vector $\boldsymbol{u}$. An illustrative example would be two control effectors with the same control effectiveness. The produced pseudo control is zero as long as both are moved symmetrically in counteracting directions.

The control allocation in INDI minimizes the control increments $\Delta \boldsymbol{u}$, not the absolute command $\boldsymbol{u}$. Therefore, it will command the actuators to the closest feasible point in the control space, which fulfills the desired pseudo control increment $\Delta \boldsymbol{v}_{\mathrm{des}}$. For example, a zero desired total pseudo control $\boldsymbol{v}_{\mathrm{des}} = \boldsymbol{0}$ does not necessarily lead to $\boldsymbol{u} = \boldsymbol{0}$ as it would be for non-incremental controllers. This effect may lead to, for example, higher energy consumption or reduced capabilities when close to the boundary of the AMS. 38,109

One solution to the path dependency problem is, for example, the restoring method by Ref. 38,109, which actively drives the absolute command to a desired value. To achieve this, an additional command in the nullspace is generated to bring the effectors to the desired value. The nullspace transition method by Ref. 17,107 has the same effect but further allows the introduction of additional constraints, beneficial for eVTOL control. Further, multiple also nonlinear secondary objectives can be considered using the approach by Ref. 110, which is related to the above-mentioned nullspace transition. Alternatively, one can use optimization-based control allocation algorithms that incorporate additional constraints.[104,111]

The control allocation includes the only part where model knowledge is required for ideal sensor-based INDI. Adaptive control allocation methods can be used to compensate for model uncertainties or increase fault tolerance. For INDI controllers, approaches adapting a control effectiveness vector [112–114] and ones adapting the full input matrix have been developed.[115,116]

While control allocation deals with distributing control efforts across available actuators, the overall success of INDI also strongly relies on the resulting stability and robustness. Ensuring stability and robustness allows the maintenance of reliable performance under different operational conditions, uncertainties, and disturbances. Therefore, the following section will summarize the extensive research efforts to investigate and enhance the stability and robustness of INDI controllers.

9. Stability and robustness

The stability and robustness of INDI controllers have attracted significant research attention, as they ensure that performance can be maintained under various conditions, including disturbances, actuator dynamics, and model uncertainties. This section reviews key studies that have contributed to our understanding of achieving and quantifying stability and robustness in INDI controllers.

The publications discussed here were selected because they address the central challenges of stability and robustness in INDI control. Each study provides valuable insights or approaches. Some of these studies offer formal stability proofs for specific INDI configurations, while others introduce methods for robustness testing or analyze the effects of practical considerations like time delays and actuator dynamics. Together, these studies provide a picture of the current state of research in this area. This section is organized as follows:

(1) Section 9.1: Discusses foundational stability analyses for MIMO and SISO systems, highlighting the effects of uncertainties, disturbances, and system dynamics.
(2) Section 9.2: Focuses on the impact of practical implementation factors such as sampling time, actuator dynamics, and time delays. It covers methods for enhancing stability margins and robustness, such as the synchronization technique. The gained insights are mainly based on numerical analysis with example systems.
(3) Section 9.3: Explores more advanced analytical approaches, robustness quantification methods, and metrics.

9.1. Foundational stability analyses

In 2019, Ref. 117 analyzed the stability of INDI control analytically for input affine MIMO systems with the same number of inputs and outputs, and with relative degree r larger than one. Their work offers stability proofs for INDI under conditions that account for actuators, sensors, and disturbances. They provide stability proofs for INDI applied to the aforementioned type of systems, using Lyapunov methods and nonlinear perturbation theory, taking into account cases that include: (A) state-dependent terms; (B) internal dynamics; (C) the presence of external disturbances; (D) regular perturbations, that are perturbations that do not change the order of the nominal system, like parametric variations; (E) singular perturbations, that are perturbations that increase the order of the system, e.g., actuator dynamics, higher order elastic dynamics.

First, using Lyapunov's theory, they derive how the state-dependent terms impact the closed-loop stability under the assumption of ideal actuators and perfect sensing without disturbances. They demonstrate that the norm of the perturbation term, $\boldsymbol{\delta}(\boldsymbol{x}, \Delta t)$, which deviates the closed-loop output

derivative from the desired pseudo control ($\boldsymbol{y}^{(r)} = \boldsymbol{v}_{\text{des}} + \boldsymbol{\delta}(\boldsymbol{x}, \Delta t)$), approaches zero as the sampling interval Δt approaches zero, meaning that the perturbation diminishes with increasing sampling frequency.

Secondly, they derive the ultimate bounds of the internal and external states for closed-loop stability under perturbations of external disturbances.

They show that INDI has better disturbance rejection than NDI by deriving that in INDI, only a disturbance increment perturbs the external dynamics. When the sampling frequency is high enough, the disturbance increments will reduce. This can be intuitively explained by the fact that part of the disturbance effects are captured in the measurement of v and compensated for by the controller.

The internal dynamics are not controlled by the INDI law. However, they are essential for the stability of the closed-loop system. Whether the internal dynamics are input-to-state stable or if only the origin of the zero dynamics is exponentially stable will impose certain limitations on the perturbations and on the initial state. They establish the conditions on initial states and disturbance magnitudes required to ensure that the states of the closed-loop system remain bounded.

When considering regular perturbations, that is, perturbations that do not change the order of the nominal system like parametric variations, they investigate the resulting perturbation terms in the NDI and INDI closed-loop system. These perturbation terms describe the difference between the nominal closed-loop external state derivatives, in particular, and the ones resulting if the regular perturbations are introduced. They show that under INDI, the perturbation terms in the closed-loop system are lower than those obtained with NDI control. They also show that the perturbing terms can be decreased even more by reducing the sampling time Δt. For singular perturbations like actuator dynamics, they show via simulation with a numerical example that decreasing Δt (increasing the sampling frequency) or increasing the gains has the benefit that the ultimate bounds decrease.

In 2022, Ref. 118 analyzes the stability and performance robustness of INDI- based control laws using the structured singular value framework.

First, they perform an analytical comparison of NDI and INDI. They consider an input affine system with a relative degree of one. They formulate the actual output dynamics of the system in terms of the model representation and additive regular perturbation terms. The system input is formulated so that the control law-generated input is affected by perturbations, which represent a class of unmodeled or neglected dynamics that occur in series connection with the system, such as actuators or high-order structural modes. The resulting $\dot{y}$ in closed-loop corresponds to v_{des} and an additive residual term that is different for NDI and INDI. They show that for an invertible control effectiveness matrix, the upper bound on the L_2 -norm of the residual term goes to zero if u and $\dot{y}$ can be measured precisely.

Their work confirms that INDI offers improved robustness to regular perturbations for the considered type of systems compared to NDI, where the residuals might grow to large amplitudes. However, it also highlights potential challenges with finite sampling times and singular perturbations such as actuator dynamics. In practice, where actuator dynamics and finite sampling times are present, they might cause the residual to grow unbounded for specific combinations of singular perturbations.

They introduce conditions to ensure boundedness and stability and investigate augmentations, such as synchronization, complementary filtering, and optimization of asynchronous designs. The authors further conduct a μ analysis of the linear version, Incremental Dynamic Inversion (IDI).

In 2022, Cunis [119] proposes a method to estimate the domain of stability or region of attraction for INDI control laws, taking the state-dependent error term into account. The considered system has a relative degree larger or equal to one, and internal dynamics exist.

Cunis proposes in his method to verify local stability by using the co-space of internal and external dynamics and locally approximating the state-dependent error term by a finite-horizon output gain. Based on the over-approximated error term and the interconnection of internal and external dynamics, the inner estimate of the closed-loop region of attraction is derived using the small-gain theorem.

9.2. Practical implementation factors from analyzing example systems

In 2018, Ref. 83 analyzed the stability and performance of an INDI law. The considered system is a SISO first-order linear system with a first-order actuator. A proportional error controller generates the pseudo control. The study has several practical implications, as the analysis focuses on the influence of: (A) controller sampling time, (B) actuator time constant, (C) time delay, (D) control effectiveness on closed-loop stability. They show that a decreasing sample time and a decreasing actuator time constant (faster actuator) both positively affect the closed-loop stability. Further, they investigate the stable region of control effectiveness uncertainty over sampling time. They conclude that for low sampling time, higher uncertainties can be tolerated. With increasing sampling time, the uncertainties that can be tolerated until the system becomes unstable decrease.

The time delay analysis examines the stable regions for a varying actuator time constant in dependency of: (A) a varying pseudo control feedback delay, (B) a varying actuator feedback delay, (C) a varying delay that is equal for both the pseudo control and actuator measurement. They show that the tolerance towards actuator delay is higher than towards a delay of the state derivative measurement. An important observation is that for the considered SISO systems, the largest stable region is obtained if both measurements, actuator, and pseudo control are delayed equally, that is, synchronized. The practical implication is that synchronizing delays in actuator and pseudo control measurements can significantly improve stability. Note that if the considered system is an overactuated MIMO system, with more actuators than pseudo controls, or if different delays are present in the different channels, then synchronization may not be as simple anymore.

Using a second example system, Ref. 83 analyzed an INDI angular rate inner-loop and an attitude outer-loop. A PI controller augments the calculation of the desired pseudo control

of the INDI controller to compensate for bias in the actuator measurement. Furthermore, they augment the controller with PCH to avoid the windup of the PI controller. They observe that unsynchronized delays between the actuator and pseudo control measurement cause an oscillatory behavior, which confirms the results obtained in the previous simple SISO example. Furthermore, they measure the influence of actuator and pseudo control measurement delays on the performance by the Root Mean Square (RMS) tracking error. They show that the lowest RMS tracking error is obtained if the delays are similar, that is synchronized.

In 2012, Ref. 49 considered a SISO linear plant and controller structure that is more complex as it includes the following characteristics:

(1) E-INDI control law.[40]
(2) first-order linear reference model with PCH.
(3) first-order differentiation filter see in Section 6.2.
(4) linear first-order sensor dynamics and transport delay on feedback signals and actuator measurements.
(5) linear first-order actuator dynamics with delay.

Their work helps to understand how practical implementation factors, such as controller gains and synchronization, influence stability margins. For the above system, Ref. 49 derived the analytical closed-loop and open-loop transfer functions. They use these transfer functions to analytically investigate the influence of the time delays (using a second-order Padé approximation), the controller gains, sensor dynamics, and PCH on stability and stability margins. They show that synchronizing the actuator measurement path with the sensor dynamics increases the stability margins.

For a SISO example, which is the roll motion for an aircraft, they determine the closed-loop stability area in dependence of the pseudo control feedback signal sensor delay τ_1 and actuator measurement delay τ_2, as depicted in Fig. 22.[49] The Figures show for two different gain values K_P for which combinations of τ_1 and τ_2 the closed-loop system is stable or not and allow the following conclusions:

(1) For small values of the sensor delay τ_1, the system remains stable even for large values of the actuator delay. In other words, the stability is more susceptible to sensor delays than actuator delays.

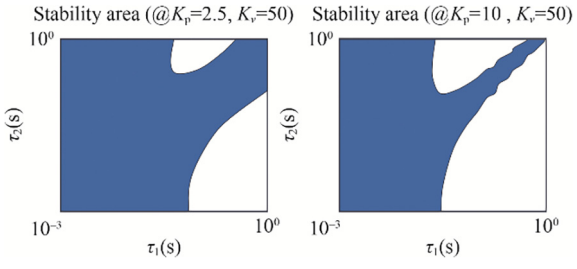


Fig. 22 Stability area (= blue) of τ_1 and τ_2 for the incremental controller with different gains qualitatively adapted from Ref. 49.

(2) If the sensor delay is large, then the system remains stable if the actuator delay is similar to the sensor delay, that is, if the two paths are synchronized. This is evident from the stable (= blue) region along the angle bisector at large delays.
(3) Increasing the controller gains will contract the stability region and impair the capability to achieve stability over synchronization.

The first two conclusions confirm the results obtained previously by Ref. 83. Based on analytically derived closed-loop transfer functions for the example system that they investigate, they further analyze the influence of PCH on command tracking, disturbance rejection, and noise suppression.

Finally, based on a numerical example using a short-period model of an aircraft, the influence of various parameters on different metrics is evaluated. The metrics considered are:

(1) Gain, phase, and time delay margins, to evaluate stability robustness.
(2) RMS value of the tracking error due to disturbances and noise to evaluate tracking performance, disturbance rejection, and noise attenuation.
(3) RMS value of the input signal, to evaluate control effort.
(4) Standard deviation of the RMS tracking error over 100 samples of uncertain plant model parameters, to evaluate performance robustness against model uncertainties.

Further, they visualize how these metrics change with increasing controller gains, control effectiveness scaling, actuator natural frequency, sensor time constant, filter bandwidth, and delays.

In 2021, Ref. 16 investigated the stability margins for their Hybrid INDI law, see also in Section 6.3, where they use a linear blend between a model-based and a measurement-based pseudo control variable. The findings are particularly relevant for systems where structural dynamics could interfere with control performance. They investigate the influence of structural vibrations and the blending factor (how much feedback is used) on the margins. Thereby, they compare several cases:

(1) With or without an equivalent low-order synchronization filter on the actuator feedback. This filter synchronizes the relative phase between actuator feedback and pseudo control feedback. Higher-order dynamics such as the sensor model, differentiation filter, and structural coupling filters might cause this phase difference.
(2) Usage of delayed u_{cmd} or u_{meas} as actuator feedback see Figs. 12 (a) and (b).

They show that for their example system under nominal conditions, that is, a nominal plant model without parametric variations, the stability margins increase if (A) Feedback in the blending is reduced (more model-based pseudo control is used), (B) Synchronization is used.

In case the assumed model in the controller is inaccurate, it is expected to be more beneficial to rely more on the measurements and feedback for robust performance.

Furthermore, they emulate structural coupling tests by adding airframe structural vibration content obtained from ground tests to the measurements. These investigations show that better stability margins under structural vibrations are obtained if instead of $u_{\text{meas}}, u_{\text{cmd}}$ is fed back (see Fig. 12)) and synchronized. The reason is that this way, structural vibrations of the control surfaces are prevented from entering the control law, which would be present in the u_{meas} measurement signal.

In 2022, Ref. 23 analyzed the impact on the stability of an Input Scaling Gain (ISG) in combination with a pseudo control estimate obtained by a second-order differentiator filter, see in Section 6.2. The study provides insights into how ISG can mitigate the adverse effects of delays and noise in the feedback signals, though with some trade-offs in tracking bandwidth.

First, they derive the closed-loop transfer function in the Laplace domain from desired to measured pseudo control for an example system given by

$$\dot{x} = x + g_0 \Delta u \tag{64}$$

with $y = x$. The measured pseudo control is obtained via a second-order differentiator filter and is subjected to additional time delays. The resulting closed-loop transfer function from desired to true pseudo control for the INDI law is a time-invariant system with delay and is also shown in Ref. 120 to represent a so-called retarded dynamic system. Retarded Functional Differential Equations (RFDE) are common in the context of time delay systems, for example, in process control, as biological and chemical processes often have to deal with large time delays.

Using a Padé approximation, they derive a stability condition based on the resulting characteristic equation by requiring the poles to be stable. The stability condition shows that the ISG increases, for the investigated system, the robustness to delays in the feedback.

Besides that, the ISG attenuates high-frequency disturbances in the pseudo control. They show that for small sampling times, the closed-loop behavior from desired to measured pseudo control corresponds to first-order dynamics, and the ISG can shape the resulting bandwidth. If the pseudo control cannot be directly measured and a differentiator is applied, the differentiation amplifies the high-frequency noise on the measured signal (compare with Section 6.1). Using a second-order roll-off filter attenuates the noise amplification caused by the differentiation (compare with Section 6.2). The ISG can further reduce the noise propagation through the controller by reducing the bandwidth of the v loop.

Furthermore, a large actuator bandwidth will result in high sensitivity to measurement noise on the pseudo control estimate, leading possibly to oscillations. The input scaling gain can reduce this sensitivity. However, as a compromise, the tracking bandwidth will be reduced as a consequence, as already mentioned.

They perform a numerical analysis for the used example system. Step and frequency responses show that the ISG can reduce the overshoot and oscillatory behavior induced by the phase lag of the differentiator filter. However, it will, at the same time, increase the settling time and decrease the tracking bandwidth.

Finally, the INDI law with ISG and second-order differentiation filter is applied to control the lateral motion of an airship. The results obtained using a nonlinear 6DoF model are shown to agree with the analysis performed on the simple example system.

9.3. Stability and robustness quantification metrics for INDI

In 2021, Seiferth [121] analyses the stability and robustness of an INDI attitude controller for an eVTOL. The nonlinear controller is trimmed and linearized. The resulting linear state-space representation of the controller is used to calculate the linear closed-loop system. He chooses analysis points at the actuators, i.e., at the actuator command signals u_{cmd}, the sensors, that is, at the feedback signals, and before the control allocation, that is, at the desired pseudo control increment. He evaluates the following metrics over an envelope given by a grid of vertical and lateral velocities:

- **SISO margins**
  (1) Nichols charts and whether the Nichols diamonds are crossed.
  (2) Gain and phase margin.

- **MIMO margins**
  (1) Maximum singular values of the complementary sensitivity function, the sensitivity function, and the balanced sensitivity function over the frequency ω are calculated at the three different analysis point locations (actuator, control allocation, sensor).
  (2) The maximal singular values over the given frequency range are used to calculate the three MIMO margins (upper + lower gain margin and phase margin).

- **μ-analysis**
  Considering the Handling Quality (HQ) Level 2 threshold or boundary from the ADS-33 standard.
- **Probabilistic μ-analysis**
  (1) Stochastic uncertainties are considered.
  (2) A lower bound value of the probability of achieving stability or a specified handling quality threshold is determined.

Contour plots over the flight envelope for the SISO and MIMO margins reveal where requirements are unsatisfied. He compares the results of the respective stability and robustness criteria.

He finds the obtained SISO margins at the actuator commands misleading, as the investigated eVTOL demonstrator has eight command channels in parallel, and instead of occurring in a single channel only, the uncertainties are more likely to occur simultaneously in these eight channels. MIMO margins, in contrast, take uncertainties simultaneously into account and are thus the more appropriate choice for the actuators.

Furthermore, the results show that the actuator cuts are less restrictive and are satisfied over a larger envelope than the control allocation cuts. The analysis point at the control allocation is a bottleneck cut: it has, compared to the actuator analysis

point, a lower number of parallel signals, making SISO margins more meaningful. As the pseudo controls represent commands to change one specific degree of freedom of the eVTOL, the respective SISO margins can be interpreted as the maximum gain and phase the controller can compensate for this specific degree of freedom. In other words, the margins can identify robustness weaknesses for specific motions.

However, the contour plots of the control allocation and the sensor analysis point match at most trim points, possibly because the number of channels at the sensor and the control allocation analysis point are similar, and a physical analogy exists. The author consequently points out that if robustness margin requirements are formulated in the future for eVTOLs, the respective location and the corresponding number of channels in parallel shall be additionally detailed in the requirement.

In 2022, Ref. 120 introduced new methods for quantifying stability and robustness in INDI systems. They introduce "parameter uncertainty bound" as a stability margin, providing a way to quantify how much model parameters can deviate from their actual values while maintaining stability. The study also links INDI stability to Neutral Functional Differential Equations (NFDE), offering a theoretical framework for analysis. First, they address stability and robustness quantification with regard to: (A) model uncertainties in control effectiveness, (B) actuator dynamics, and (C) controller sampling rate. They consider input affine MIMO systems with an equal number of inputs and outputs. All outputs are assumed to have a relative degree of one, and no internal dynamics exist.

Concerning (A) model uncertainties in the control effectiveness and (B) the presence of actuator dynamics, they derive analytically the values of the time delay margin, and "parameter uncertainty bound" stability condition.

They obtain a stability condition for the controller sampling rate from the time delay margin.

The "parameter uncertainty bound" is a stability margin definition that they introduce to quantify the stability condition of INDI laws. For SISO systems, it corresponds to the ratio of the actual control effectiveness B_v and a specific control effectiveness value $\widehat{B}_v$ used in the control law: Its value represents the largest tolerable deviation of model parameters from the actual value that still maintains a stable closed-loop system.

A finding that allows them to derive the stability conditions is that the closed-loop INDI can be shown: (A) to correspond to a special class of neutral functional differential equations (NFDE) if model uncertainties in the control effectivity exist, (B) to correspond to high-order RFDE in the presence of actuator dynamics. NFDE is common in the context of time-delay systems. These are particularly used in process control because biological and chemical processes often inherit considerable time delays. By taking the Laplace transform of the closed-loop NFDE, the system's poles can be calculated, and conditions for the stability can be derived for the allowed uncertainty and time delay margin. Thereby, they use the results provided in the literature with regard to the calculation of time-delay margins for time-delay systems.

Finally, they find that underestimating the control effectiveness in the INDI control law improves the performance. An intuitive interpretation is that underestimating (= decreasing) the control effectiveness corresponds to increasing the gain of the controller because of the inversion of the control effectiveness in the control allocation.

10. Gain design for INDI

Ref. 80 presented a gain design process for an INDI-based flight control algorithm. Ref. 85 extended this strategy to actively consider the various filters used within the INDI controller in the gain design process. Another extension they make is to incorporate disturbance rejection requirements into the gain design. The gain design uses MATLAB/ Simulink with systune and linear plant and controller models. They trim and linearize the plant and controller separately. They set analysis points and tunable gains, which impose requirements over the Tuning Goals of the error controller gain tuning routine. The considered requirements are tracking in case of a step command, disturbance rejection in case of step disturbances on feedback signals, a constraint limiting the controller gain values, and gain and phase margins. They consider two design cases: hover and wing-borne flight. The controller structure is identical for both phases, but the gains differ. During transition flight, the gains fade with a blending that depends on current airspeed. They assess the gain design with a nonlinear model, including noise and delays.

Ref. 88 presented a parameter tuning process applied to an incremental nonlinear dynamic inversion controller, where both, controller gains and filter time constants, are optimized. They formulate a multiobjective optimization problem based on various requirements. The incorporated requirements are:

(1) Time Domain Reference Tracking: Incorporated into the cost function over the Integral of the Absolute Error (IAE) for the tracking error and its derivative.
(2) Disturbance Rejection: minimizing the maximum magnitude of the frequency response of the respective linear disturbance transfer function.
(3) Stability margins: Imposed by a Nichols requirement, standard SISO phase, and gain margin and disc margins.
(4) Lead-lag Filter Magnitude: A frequency-dependent limiting of the magnitude of a lead-lag filter is imposed to prevent the amplification of high-frequency disturbances like vibrations.

The lead-lag filter, introduced at a dedicated position within the control structure, introduces an additional relative degree of freedom, as shown analytically. This additional degree of freedom allows them to meet all requirements, especially the Nichols stability requirement, by allowing different shapes of the open-loop frequency response.

Rather than relying solely on linear models, the optimization process uses a high-fidelity nonlinear model. This allows them to account for nonlinear effects like sensor errors, vibrations, and absolute and rate limits within the optimization and parameter design. This approach provides a more suitable solution for real-world applications. Moreover, the multiple robustness metrics considered in their approach, especially those tailored for MIMO systems, aim to provide a sufficient level of robustness given the novelty of the vehicle and control strategy. Finally, they assess the resulting controller in both simulation and flight tests.

Ref. 122 also employed a gain tuning approach that actively incorporates the nonlinear closed-loop dynamics. They present a multiobjective gain design method that accounts for both nonlinear closed-loop dynamics to meet robustness and performance requirements and the linearized model to address stability and robustness in the linear domain.

By inherently including nonlinear dynamics in the optimization process, the method ensures a more accurate representation of the actual closed-loop behavior compared to relying solely on linearized models. This reduces the need for extensive gain design-verification iterations.

The Kreisselmeier-Steinhauser vector performance index combines pattern-search optimization to integrate the nonlinear model's requirements with linear stability requirements. Throughout the optimization process, conditional relinearization maintains consistency between the linear and nonlinear systems.

They demonstrate the approach on an INDI control law for a 3-ton class, manned, electric Vertical Take-Off and Landing (eVTOL) vehicle.

To demonstrate the advantages of incorporating the nonlinear model, the results were compared with gains generated by systune, using only the linear model. The comparison focused on meeting the requirements. The proposed approach showed benefits, particularly in suppressing undesired axis coupling and accounting for actuator/motor limits. It also offered faster turnaround times, especially with frequent plant or controller updates, without needing manual adjustments to gain limits. Nonlinear effects, especially for higher-amplitude inputs, are not visible in linear models, making the proposed method more accurate. Additionally, the proposed approach provides greater flexibility, whereas systune often requires workarounds due to its more restrictive tuning goals.

11. Conclusions and outlook

This survey provides an overview of modern trends in INDI, focusing on its various components. In addition to the INDI control law, an INDI-based controller consists of several components, such as control allocation, reference models, error controllers, and state and pseudo control estimation.

Each of these components is reviewed in detail, comparing different approaches and bringing them into context with each other. Key challenges are highlighted.

Several possibilities are explored to obtain the desired pseudo control command, such as deriving it from outer cascaded control loops or using reference models in combination with an error controller. Several reference model designs are presented, addressing their respective advantages and features.

For example, linear models offer simplicity, while cascaded physical reference models, though still relatively simple, incorporate limits, rate saturations, and inner-loop or actuator dynamics to generate physically feasible commands. Higher-order reference models, especially in the context of E-INDI, can provide feedforward signals that account for state-dependent terms, improving the performance. Controlled linear incremental physical plant models, can provide feasible reference signals across multiple variables. This is beneficial if adjusting control variables or blending them with operating or input conditions is required.

Multiple possibilities exist regarding the feedback of the control input. For example, the u feedback signal can be derived from a direct measurement, an actuator model, an estimator, or a delayed command signal. For sufficiently fast actuators, a delayed command signal can yield good results. Using the measured plant inputs as feedback can introduce issues due to noise or structural vibrations that are contained in the measured u signal and may reduce stability margins. The control input command tracking can sometimes be deteriorated by backlash effects. An interesting question that might be addressed in the future is how these effects might impact INDI-based control and what measures would be necessary to mitigate them.

Regarding the methods for obtaining the pseudo control variable, that have been thoroughly investigated in the literature, derivative filters, complementary filters, extended state observers, and other techniques are presented. Different complementary filters, combining measurement and model-based signals, are compared and their similarities to an approach that uses an extended state observer are discussed. For instance, the complementary filter corresponding to the extended filter approach shows better noise attenuation in the frequency domain due to a higher roll-off at high frequencies.

Further, this review addresses challenges such as synchronization in feedback and actuator command paths, summarizing relevant publications that tackle these issues.

Control allocation is a crucial component, particularly for overactuated systems with more inputs than pseudo controls. This paper focuses on challenges in control allocation relevant to INDI, emphasizing pseudo-inverse-based approaches. Challenges such as path dependency are addressed with various strategies like the restoring method and nullspace transition.

Extended INDI, which incorporates actuator dynamics, directs towards future. Future approaches may aim for a unified control framework, combining elements of the E-INDI method with actuator control. In particular, the actuator's intrinsic input feedback (from the actuator plus its controller) theoretically cancels with the INDI input feedback. In practice, however, this cancellation is imperfect due to sensor delays and dynamics and because the actuator and INDI are running on different computers/hardware with possibly different sampling rates. Future research can further explore these details.

Regarding stability and robustness, this survey provides an overview of some works that address foundational stability analyses, practical implementation considerations from example systems, and advanced metrics for quantifying stability and robustness. Key insights from the publications suggest that INDI, depending on the system, may offer superior disturbance rejection compared to NDI and that regular perturbations (such as parametric variations) may have less impact under INDI. Actuator dynamics can introduce notable perturbations. Reducing the sampling time and using faster actuator dynamics might improve closed-loop stability. Maintaining a high sampling frequency is generally beneficial for stability. Internal dynamics can pose risks, and disturbance intensity should be carefully considered to ensure closed-loop stability.

While actuator delays impact stability less than pseudo control feedback delays, synchronizing delays in both feedback paths can significantly enhance stability. In overactuated systems with varying delays across channels, synchronization might be more complex but remains crucial for ensuring stabil-

ity. A complementary filter and synchronization may represent a reasonable means to address this challenge. An Input Scaling Gain (ISG) can enhance robustness against feedback delays, though it may decrease tracking bandwidth.

In overactuated systems, MIMO margins might be more appropriate for actuator commands due to the simultaneous presence of uncertainties across multiple channels. SISO margins may be misleading in this context, as they consider single channels in isolation. At the control allocation, that is, at the desired pseudo control increment, SISO margins can provide valuable insights as fewer signals are in parallel, and the control allocation can hence be a critical bottleneck. As such, it can provide insights into the robustness of specific degrees of freedom, highlighting potential weaknesses in the controller's ability to control individual motions. In addition, the analysis point– whether at the actuators or control allocation, plays a role in assessing stability and robustness. Future eVTOL requirements could specify MIMO margin thresholds, the specific analysis points, and the number of channels involved, as these factors affect the interpretation of robustness margins across the flight envelope.

Finally, the latest trends in the gain tuning of INDI-based control laws are to incorporate a controller implementation, which is as close as possible to the actual implementation. This means that rather than using purely linear models the gain tuning routines include the controller with all its nonlinearities. This approach reduces the number of iterations between gain design and assessment compared to using only a linear model for tuning and then assessing with the nonlinear model. By using the nonlinear model during tuning, nonlinearities such as rate saturations, protections or nonlinear cross-couplings can be considered earlier in the process.

CRediT authorship contribution statement

Agnes STEINERT: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Conceptualization. **Stefan RAAB:** Writing – review & editing, Writing – original draft, Visualization, Validation. **Simon HAFNER:** Writing – review & editing, Writing – original draft, Visualization. **Florian HOLZAPFEL:** Writing – review & editing, Supervision, Funding acquisition. **Haichao HONG:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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